

The University of Sydney

School of Electrical and Computer Engineering

AI-Powered Business English Certification Preparation System

ELEC5620 Model-Based Software Engineering
Project Stage 1: Requirements, Architecture Design and System Modelling

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AI Acknowledgement

The project team acknowledges the use of generative AI tools in the preparation of this proposal. The tools used were:

- **Claude Sonnet 5.5** (Anthropic, `https://www.anthropic.com`), using its web search feature to identify relevant literature and its file creation feature to assist with document drafting;
- **OpenAI Codex** (GPT-5), published by OpenAI (`https://openai.com/codex/`); and
- **GPT-5.6 Sol**, published by OpenAI (`https://openai.com`).

These tools were used for the following purposes:

- to build our understanding of different types of diagrams, and to propose search phrases and relevant literature for the project;
- to help organise and assemble the information presented in our diagrams; and
- to assist with document structure and formatting in LaTeX, and with grammar and clarity of English expression.

The project team has reviewed and verified all diagrams, content, information, technical claims, and the final wording of this proposal. The submitted work remains the sole responsibility of the project team.

1 Introduction

1.1 Background

Large language models (LLMs) now make it possible for one person to run a company supported by specialised AI agents that plan, decide, use tools and collaborate. This project applies that "One-Person AI Company" idea to education.

Many working professionals need an English certification such as IELTS or TOEFL for university admission, visa applications or promotion. Good tutoring is expensive, inflexible and inconsistent, and few tutors give detailed feedback on all four skills (Speaking, Writing, Reading and Listening). Learners therefore follow inefficient study paths and often sit the exam more than once.

1.2 Proposed Solution

We propose an AI-powered exam preparation platform run by a single human instructor (the founder) and four specialised AI agents:

- **Learning Coordinator Agent:** onboards learners, builds their profile, creates and adapts a personalised learning plan, recommends materials and manages handoffs between agents.
- **Tutor Agent:** runs speaking, writing, reading and listening practice, generates prompts and gives immediate, detailed feedback.
- **Assessment & Analytics Agent:** scores work consistently against the official rubrics, identifies weaknesses and error patterns, and predicts exam readiness.
- **Operations & Quality Manager:** checks that agent outputs are consistent and confident, detects curriculum gaps and escalates problem cases to the human instructor.

Company brief. *For working professionals preparing for IELTS or TOEFL, reliable feedback on all four skills is expensive, slow and inconsistent because it depends on human tutors. Our one-person AI company coordinates a Learning Coordinator, a Tutor, an Assessment & Analytics agent and an Operations & Quality Manager to deliver personalised, quality-checked exam preparation, while the human founder approves low-confidence or contradictory results.*

1.3 Report Structure

Section 2 presents the requirements and their classification, including each member's ad hoc requirement and the feature diagram. Section 3 gives the use case model and specifications. Section 4 describes the architecture and design rationale. Section 5 presents the class, object and collaboration diagrams, and Section 6 each member's behaviour models. Sections 7 to 9 cover the role of the LLM, assumptions, and change assessment and acceptance. Individual contributions are listed in Section 10.

2 Requirements

2.1 Stakeholders and Actors

The agents are part of the system, so they are not actors. The external actors are:

Actor	Role and goals
Learner	A working professional preparing for IELTS or TOEFL. Wants targeted practice, fast and trustworthy feedback, and a realistic estimate of their band score.
Human Instructor / Founder	Runs the company. Supervises the agents, reviews escalated cases, sets quality thresholds and coaches learners who are struggling.
LLM Service (external system)	Third-party large language model accessed through an API. Generates prompts, feedback, scores, consistency verdicts and recommendations.
System Clock (timer)	Triggers scheduled behaviour, such as weekly learning plan adaptation.

Table 1: External actors.

2.2 Ad Hoc Requirements

Each member contributed at least one ad hoc requirement: an informal description written from a stakeholder's point of view. As taught in Tutorial 3, ad hoc requirements mix *what* the system does, *how* it does it and *how well* it must do it. Each member's use case and behaviour models (Sections 3 and 6) are derived from their own ad hoc requirement.

2.2.1 Practising an IELTS/TOEFL Essay and Getting Feedback I Can Trust (Naratorn Pisedtasalasai)

Scenario. Mai is an accountant who needs IELTS band 7 to apply for a job overseas. Writing is her weakest skill, and a private tutor who marks essays costs too much and takes days to reply. She wants to practise a Task 2 essay tonight and know straight away what she is doing wrong.

She asks the platform for a practice question and gets one in the same style as the real exam. She has 40 minutes to write about 250 words. After she submits, comments start appearing within a few seconds: a wrong tense, a vague topic sentence, the word "important" used five times. Each comment says why it is a problem and shows a better version. She also sees an estimated band for each of the four writing criteria.

Mai never sees a result that contradicts itself. Behind the scenes, the system compares the comments with the band scores. If they do not match, or the AI is not sure of its marking, the essay goes to the human instructor first, and Mai is told it is being double-checked. Once she has her feedback, she rewrites the weak paragraphs and submits again, and the system shows her what improved since her first draft.

Ad hoc requirements.

Getting a question

1. The learner can ask for a Task 2 (or TOEFL independent) essay question at any time. *(what)*
2. The question is written by the LLM, using the learner's target exam and current writing band so it is neither too easy nor too hard. *(how)*
3. A timer and word counter are shown while writing, as in the real test. *(what)*

Getting feedback

4. After submission, the learner gets comments on tense and grammar errors, repeated or informal vocabulary, weak topic sentences and poor paragraph linking. *(what)*
5. Every comment points to the exact sentence, says why it is wrong and gives an improved version. *(what)*
6. The learner should not be left staring at a blank screen: comments start appearing within a few seconds, instead of waiting for the full critique to finish. *(how well)*
7. The learner also gets an estimated band for each of the four writing criteria (Task Response, Coherence & Cohesion, Lexical Resource, Grammatical Range & Accuracy) and an overall band. *(what)*

Checking the result before the learner sees it

8. The LLM returns the bands in a fixed structure, with a "how sure am I" score for each one, so the result can be checked automatically. *(how)*
9. The system compares the comments with the bands. For example, if the comments praise the vocabulary but Lexical Resource is 5.0, that is a mismatch. *(what)*
10. If there is a mismatch, or the AI is less than 70% sure, the result is held back and goes to the human instructor. The learner sees "your essay is being double-checked". *(what / how)*
11. The instructor should not be flooded with false alarms (about 1 in 50 at most), and real mismatches should almost never get through to the learner. *(how well)*

Improving the essay

12. The learner can edit the same essay and resubmit it without starting from scratch. *(what)*
13. The second round of feedback says which earlier problems have been fixed and which are still there, and shows the band change for each criterion. *(what)*
14. Most learners should find the feedback clear enough to act on without asking anyone else. *(how well)*

Getting study materials

15. After feedback, the learner can ask for study materials that target the mistakes found, for example a lesson on academic synonyms when the same word is overused. *(what)*
16. The LLM ranks the materials and writes a one-line reason for each, and the recommendations go through the same quality check as the feedback. *(how)*
17. The recommendations should load within a few seconds, and a learner with no results yet still gets sensible starter materials. *(how well)*

Where the LLM fits. Following the agent model from Tutorial 1:

Element	**In this scenario**
Environment	Mai's profile and target band, her essay drafts, the official writing band descriptors [1], the instructor
Perception	The LLM reads the essay and picks out grammar, vocabulary and structure problems
Decision-making	The LLM estimates the bands; the system decides whether to release the result or send it for a double-check
Action & tools	Write the question, show the comments and bands, alert the instructor, compare drafts
Feedback	Resubmitted drafts and instructor corrections show whether the AI marked fairly
Human oversight	The instructor checks any result that contradicts itself or where the AI is unsure

Table 2: Agent elements for the essay-feedback requirement.

2.2.2 Registration and Profile Creating for users preparing for an English level exam (Menglu Lin)

Scenario. Alex is a working professional preparing for IELTS. He has never used the platform before and does not know which skills he should prioritize. He wants to create an account, tell the system about his current English level, target score, available study time and exam date, and immediately receive a study plan that fits his circumstances.

Ad hoc requirements.

Creating an account and learner profile

1. The learner can register by providing the required account information and basic learning information such as target exam, current level, target score and expected

exam date. *(what)*

2. The system checks that all mandatory information is present and valid before creating the learner profile. *(how)*
3. If information is missing or invalid, the learner is told exactly which field needs to be corrected instead of being asked to complete the whole form again. *(what / how)*
4. The learner can review the generated profile before confirming it and can modify incorrect information. *(what)*
5. Profile creation and updates should complete quickly enough that the learner does not notice a significant delay. *(how well)*

   Creating the first learning plan

6. Once the learner confirms the profile, the system can generate a personalised learning plan based on the learner's target exam, target score, current level, weaknesses and available study time. *(what)*
7. The Learning Coordinator prioritises skills that have the largest gap between the learner's current level and target level. *(how)*
8. The generated plan explains why each skill or activity has been given its priority, so the learner understands why the system is recommending it. *(what / how)*
9. The learner can accept the proposed plan or request changes before it becomes the active learning plan. *(what)*
10. If there is not enough profile information to create a reliable plan, the system asks the learner for the missing information rather than generating an arbitrary plan. *(how)*
11. The initial learning plan should be available soon after the learner confirms their profile. *(how well)*
12. Most learners should be able to understand what they need to study next without requiring help from the human instructor. *(how well)*

Where the LLM fits. Following the agent model from Tutorial 1:

Element	In this scenario
Environment	The learner's registration information, learner profile, target exam, target score, current proficiency level, study availability and learning goals
Perception	The LLM interprets the learner's goals, current level, constraints and preferences from the confirmed learner profile
Decision-making	The LLM helps the Learning Coordinator determine which skills should be prioritised and how the initial personalised learning plan should be structured
Action & tools	Generate the initial learning plan, explain the rationale for each priority, and suggest appropriate study activities or materials
Feedback	Learner acceptance, requested changes and later assessment results indicate whether the plan was appropriate and provide information for future plan adaptation
Human oversight	The human instructor reviews cases where the learner profile is incomplete, the generated plan is inconsistent with the learner's goals, or the AI result has low confidence

Table 3: Agent elements for the learner onboarding and personalised learning plan requirement.

2.2.3 Taking a Mock Examination and Understanding My Exam Readiness

(Yuxuan Tang)

Scenario. Kit is a working professional preparing for IELTS. She has completed several practice activities but is still unsure whether she is ready for the real examination. She wants to take a full mock examination covering Listening, Reading, Writing and Speaking and receive an overall estimate of her current performance.

Kit starts the mock examination and completes all four sections. After submission, the system produces skill scores, an estimated overall band and an indication of her exam readiness. It also compares her latest result with previous assessments so she can see which skills are improving and which weaknesses continue to appear.

Kit then opens her Performance Report to review her strengths, weaknesses and progress over time. If an important assessment result is uncertain or inconsistent, it is reviewed before being treated as final. The report helps Kit understand whether she is approaching her target score and what she should focus on next.

Ad hoc requirements.

Taking a mock examination

1. The learner can start a full IELTS or TOEFL mock examination covering Listening, Reading, Writing and Speaking. *(what)*
2. Before the examination starts, the system checks that the learner has a profile, learning plan and sufficient preparation history. *(how)*
3. If the learner does not complete all required sections, the examination is recorded as incomplete and no complete final result is shown. *(what)*
4. The learner's progress is saved during the examination so completed work is not unnecessarily lost after an interruption. *(how)*

   Scoring and exam readiness

5. After completion, the system produces results for all four skills and an estimated overall IELTS band or TOEFL score. *(what)*
6. The system uses the mock-examination result to estimate the learner's readiness for the real examination. *(what)*
7. Scoring should remain consistent when responses are assessed against the same rubric, with inter-rater reliability above 0.85. *(how well)*
8. Scoring and initial report generation should normally complete within five seconds. *(how well)*
9. If an important result is low-confidence or inconsistent, it is not treated as final until it has been reviewed. *(how)*

   Understanding performance

10. The learner can view a Performance Report showing strengths and weaknesses across Listening, Reading, Writing and Speaking. *(what)*
11. The system compares current and previous assessment results to identify improvement trends and repeated weaknesses. *(what)*
12. If there is not enough data to determine a reliable trend, the system tells the learner that more assessment data is needed. *(how)*
13. The report provides recommendations based on the learner's main skill gaps and progress toward the target score. *(what)*
14. The report should be clear enough for most learners to understand what they should focus on next without additional help. *(how well)*

Where the LLM fits. Following the agent model from Tutorial 1:

Element	In this scenario
Environment	The learner's profile, target exam, target score, mock examination responses, previous assessment results and learning goals
Perception	The LLM interprets the learner's mock examination responses, previous results and current performance across the four skills
Decision-making	The LLM helps the Assessment & Analytics Agent determine the learner's strengths, weaknesses, estimated score and exam readiness
Action & tools	Generate the assessment result, identify performance trends, and suggest appropriate areas for further improvement
Feedback	Later practice activities and assessment results indicate whether the identified weaknesses and readiness estimate were appropriate
Human oversight	The human instructor reviews cases where the assessment result is inconsistent, uncertain or has low confidence

Table 4: Agent elements for the mock examination and performance report requirement.

2.2.4 **Spotting a Curriculum Gap Before It Wastes Everyone's Time** (Mohamed Azim)

Scenario. Grace is the human instructor who runs the platform. She does not have time to read every learner's feedback log, but she still needs to know when something in the course itself is broken, not just when one learner is struggling.

Over the past week, the Operations & Quality Manager has been quietly comparing error patterns across all active learners. It notices that fourteen different learners, at different levels and with different coaches assigned, have all been marked down for the same thing in their Writing feedback: misusing conditional sentences ("If I will go..." instead of "If I go..."). That is unusual; normally only two or three learners share a weak point at once, and the error is normally so basic that higher-band learners do not make it. Here, even a band-7 learner made the mistake twice.

The system does not just add this to a list; it checks whether the practice material itself might be teaching the wrong thing. It looks at which prompts and example sentences the Coach has been generating for conditional-sentence exercises, and finds that the LLM's own example sentences contain the same error in three of the last ten generated prompts. That is the real problem: a flawed template is quietly misleading dozens of learners, not a coincidence of individual mistakes.

Grace gets a single notification, not fourteen. It says what the pattern is, how many learners are affected, what level they are at, and, because the system checked its own material, that the root cause looks like a faulty example in the practice-generation prompt rather than the learners. She reviews the flagged examples, confirms the system's diagnosis, and fixes the template in one place. The next time a learner requests a conditional-

sentence exercise, the corrected version is used, and the Quality Manager keeps watching to confirm the error rate actually drops before it closes the case.

Ad hoc requirements.

Watching for shared patterns

1. The Operations & Quality Manager continuously aggregates error tags (e.g., "conditional sentence misuse") across all active learners, not just within one learner's history. *(what)*
2. A pattern is only flagged as a possible curriculum gap once the number of learners sharing it in a given week exceeds what is normal for that error type and learner level, not on a fixed raw count. *(how)*
3. The system distinguishes an error that is expected at a learner's level (e.g., article misuse at band 4) from one that is unusual for it (the same error appearing at band 7), and weights the second kind as more suspicious. *(how)*

Telling material errors apart from learner errors

4. When a pattern is flagged, the system checks whether the error also appears in the Coach's own recently generated practice prompts and example sentences for that topic. *(what)*
5. If the system's own generated material contains the error, the notification names the likely cause as the material rather than listing it as a learner weakness. *(what)*
6. The instructor is shown the specific flawed example sentences or prompts, not just a description of the pattern, so she does not have to go hunting for them herself. *(how)*

Keeping the instructor's attention worth something

7. Grace receives one consolidated notification per distinct pattern, never one notification per affected learner. *(what)*
8. The instructor should not be alerted more than roughly once a week for patterns that turn out to be normal variation; most notifications she receives should reflect a real, actionable issue. *(how well)*
9. Grace can approve the diagnosis, mark it as a false alarm, or request more learner examples before deciding, and her response trains which future patterns get escalated at the same sensitivity. *(what)*

Closing the loop

10. Once Grace confirms a fix (e.g., a corrected prompt template), the Quality Manager keeps tracking the same error tag for a set follow-up period to confirm the rate drops. *(what)*
11. If the error rate does not drop within that follow-up period, the case is automatically reopened and re-escalated rather than silently closing. *(how)*
12. Most flagged-and-fixed patterns should show a measurable drop in recurrence within two weeks of the fix, or the detection logic itself needs reviewing. *(how well)*

Where the LLM fits. Following the agent model from Tutorial 1:

Element	**In this scenario**
Environment	Error tags and scores from every learner's Writing/Speaking sessions, the Coach's own generated practice prompts and example sentences, and Grace as the instructor
Perception	The LLM reads aggregated error tags across learners and separately inspects the Coach's recent generated content for the same error
Decision-making	The system decides whether a shared error is ordinary variation, a cohort-wide weak spot, or a flaw in the generated material itself, and whether it clears Grace's escalation threshold
Action & tools	Aggregate and compare error logs across learners, check the flagged material against correct usage, compile one consolidated notification, and re-run the check after a fix
Feedback	Grace's approve / false-alarm / request-more-examples decision, and the post-fix error-rate trend, both tune future sensitivity
Human oversight	Grace decides whether the diagnosis is right and whether/how to fix the underlying material; the system never edits its own prompts unsupervised

Table 5: Agent elements for the curriculum-gap detection requirement.

2.2.5 Practising an IELTS/TOEFL Speaking Task and Getting Feedback We Can Trust (Suhani Saha)

Scenario. Suhani is a working professional who prepares for an IELTS or TOEFL exam. Speaking is one of her weaker skills, and she wants to practise a realistic speaking task without needing to wait for a private tutor. She asks the platform for a speaking task based on her target score and recent weaknesses.

The system provides a suitable speech prompt and gives her preparation and speech timers similar to the real examination. She records and submits her response. The system

then provides feedback on pronunciation, fluency, vocabulary, grammar, hesitations, and structure, together with estimated scores.

Before showing her the result, the system checks whether the feedback is consistent with the estimated scores. If the AI is uncertain or the feedback and scores contradict each other, the result is held and sent to the human instructor for review. Once the feedback is verified, she receives it and can re-record the same task to see whether her performance has improved.

Ad hoc requirements.

Getting a speaking task

1. The learner can request an IELTS Part 2 cue card or TOEFL independent speaking task. *(what)*
2. The speaking prompt is generated by the LLM using the learner's target exam, target band/score and recently identified weaknesses. *(how)*
3. The system provides a preparation timer and speaking timer that reflect the selected examination format. *(what)*

Getting speaking feedback

4. After submitting the recording, the learner receives feedback on pronunciation, fluency and coherence, vocabulary, grammar, filler words and hesitations. *(what)*
5. Each feedback comment identifies the relevant part of the response, explains the problem and provides an improved example where appropriate. *(what)*
6. Initial feedback should begin appearing within a few seconds rather than requiring the learner to wait for the complete analysis. *(how well)*
7. The learner receives estimated scores for Fluency & Coherence, Lexical Resource, Grammatical Range & Accuracy, and Pronunciation, together with an overall estimated band. *(what)*

Checking the result before release

8. The LLM returns the estimated scores together with a confidence score for each criterion so that the result can be checked automatically. *(how)*
9. The system compares the feedback comments with the corresponding estimated scores to identify inconsistencies. *(what)*
10. If the AI confidence is below 70%, or if the comments and scores are inconsistent, the result is held and sent to the human instructor for review. *(what / how)*
11. The system should minimise unnecessary escalations while ensuring that genuine inconsistencies are detected before feedback reaches the learner. *(how well)*

Improving the speaking attempt

12. The learner can re-record the same speaking task without starting a completely new practice session. *(what)*
13. The second feedback report identifies which previous problems have improved and shows changes in the estimated score for each criterion. *(what)*
14. The feedback should be clear enough for the learner to understand what they need to improve without requiring additional explanation from the instructor. *(how well)*

2.3 Classification of Requirements

Requirements are classified into three groups:

- **Agreement:** commitments agreed with the stakeholders that constrain the whole system and are not negotiable.
- **Mandatory capabilities:** features that must be present for the system to deliver its core value (the minimum viable product).
- **Optional features:** features that add value but can be deferred to a later release.

Class	Requirement	Agent
Agreement	Each learner prepares for exactly one target exam at a time (IELTS or TOEFL).	Coordinator
Agreement	No AI-generated feedback, score or recommendation is released to a learner without passing the quality check.	Ops & Quality
Agreement	The human instructor has the final decision on any escalated result.	Ops & Quality
Agreement	Learner data sent to the external LLM is stripped of personal details.	All agents
Mandatory	Registration and learner profile; personalised learning plan creation and adaptation; progress tracking.	Coordinator
Mandatory	Practice in at least one of Speaking, Writing, Reading and Listening, with instant AI feedback.	Tutor
Mandatory	Mock examinations, question bank, rubric-based scoring and performance reports.	Assessment

Class	Requirement	Agent
Mandatory	System health monitoring, agent consistency checks, low-confidence detection, escalation to the instructor and audit logging.	Ops & Quality
Optional	Recommended materials.	Coordinator
Optional	Readiness (band score) prediction.	Assessment
Optional	Curriculum gap detection and inactivity reminders.	Ops & Quality

Table 6: Requirement classification.

2.4 Functional Requirements

Agent	Feature	Description
Learning Coordinator	Progress Tracking	Monitor and track the learner's progress across all activities and assessments.
	Status Updates	Maintain the learner's current status (proficiency level, readiness, plan adherence).
	Profile Management	Collect the initial profile and keep it updated based on performance.
	Learning Plan Creation	Generate personalised learning paths based on the learner's goals.
	Plan Adaptation	Update and adjust the learning plan based on assessment results.
	Recommended Materials	Recommend study materials that target the learner's weaknesses, with a reason for each.
Tutor	Speaking	Conduct mock interviews, evaluate pronunciation and fluency, and provide conversation practice with real-time feedback.
	Reading	Present comprehension exercises, explain vocabulary and adjust difficulty based on performance.

Agent	Feature	Description
	Writing	Generate essay prompts, give detailed feedback on grammar and structure, and allow revision with updated critiques.
	Listening	Deliver audio materials at various speeds, assess comprehension and clarify difficult concepts.
Assessment & Analytics	Test Administration	Administer practice tests and mock exams in IELTS/TOEFL format.
	Question Bank	Maintain and select appropriate questions for each learner level.
	Accuracy Scoring	Score responses consistently and fairly against established rubrics.
	Statistics Generation	Calculate performance metrics, improvement trends and weakness analysis.
	Readiness Prediction	Estimate exam readiness and the likely band score.
	Performance Reporting	Generate reports showing strengths and weaknesses per skill.
Operations & Quality Manager	System Health Monitoring	Ensure all agents are functioning correctly and not producing errors.
	Agent Consistency Checks	Verify that results from different agents agree and do not contradict each other.
	Low-Confidence Detection	Flag decisions or recommendations whose confidence is below the threshold.
	Escalation Management	Send complex, contradictory or low-confidence cases to the human instructor.
	Curriculum Gap Detection	Detect when many learners struggle with the same topic.
	Inactivity Notifications	Remind learners who have not practised for 7 or more days.

Table 7: Functional requirements.

2.5 Non-Functional Requirements

Agent	Category	Requirement
Learning Coordinator	Performance	Profile updates complete in less than 1 second.
	Availability	Learning plans are available 99.9% of the time.
	Explainability	Plan explanations are clear and understandable to diverse learners.
	Scalability	Supports updates for 100+ concurrent learners.
Tutor	Performance	First feedback appears within 3 seconds.
	Usability	More than 85% of learners rate the feedback as helpful.
	Usability	Audio materials are clear, professional quality (bitrate above 128 kbps).
	Accuracy	Pronunciation assessment matches human evaluation more than 85% of the time.
Assessment & Analytics	Reliability	The same response receives the same score across assessments (inter-rater reliability above 0.85).
	Accuracy	Scoring is calibrated monthly against the official IELTS/TOEFL rubrics.
	Performance	Scoring and report generation complete within 5 seconds.
	Security	Assessment data is stored encrypted and retained for 7 years.
Operations & Quality Manager	Performance	Quality issues are detected within 1 minute of occurring.
	Reliability	False positive rate below 2%, false negative rate below 1%, measured against an instructor-labelled sample.
	Performance	Inactivity reminders are delivered within 24 hours.
	Auditability	All quality checks and escalations are logged.

Table 8: Non-functional requirements.

2.6 Feature Diagram

Figure 1 shows the feature model [2]. The root is the system; the first level groups features into the four capabilities delivered by the agents. Filled circles mark mandatory

features and hollow circles optional ones. *Target Exam* is an alternative (xor) group, since exactly one of IELTS and TOEFL is chosen, and *Practice Skills* is an or group, since at least one skill is practised. Non-functional requirements are attached to the features they constrain, and four cross-tree constraints record dependencies between features.

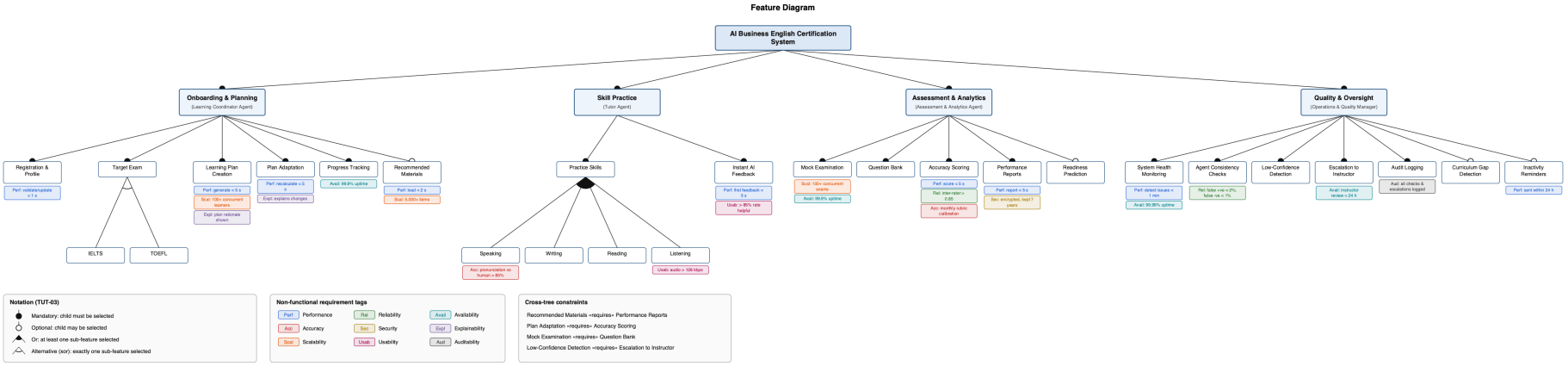


Figure 1: Feature diagram with non-functional requirements and cross-tree constraints.

3 Use Case Model

3.1 Overall Use Case Diagram

Figure 2 shows the overall use case diagram. The system boundary contains one module per agent. The external actors are the Learner, the Human Instructor / Founder, the LLM Service and the System Clock.

- **Generalisation:** the abstract use case *Complete Skill Practice* is specialised by UC3 (Speaking), UC4 (Writing), Reading and Listening practice. The children inherit its associations with the Learner and the LLM Service.
- **Include:** UC4, UC5 and UC10 all include UC9 *Check Agent Output Quality*, so every AI result is checked before release. UC4 and UC5 share *Score Response*, and UC6 and UC7 share *Analyse Performance*.
- **Extend:** UC8 *Review Escalated Case* extends UC9 at the extension point *issue detected*.

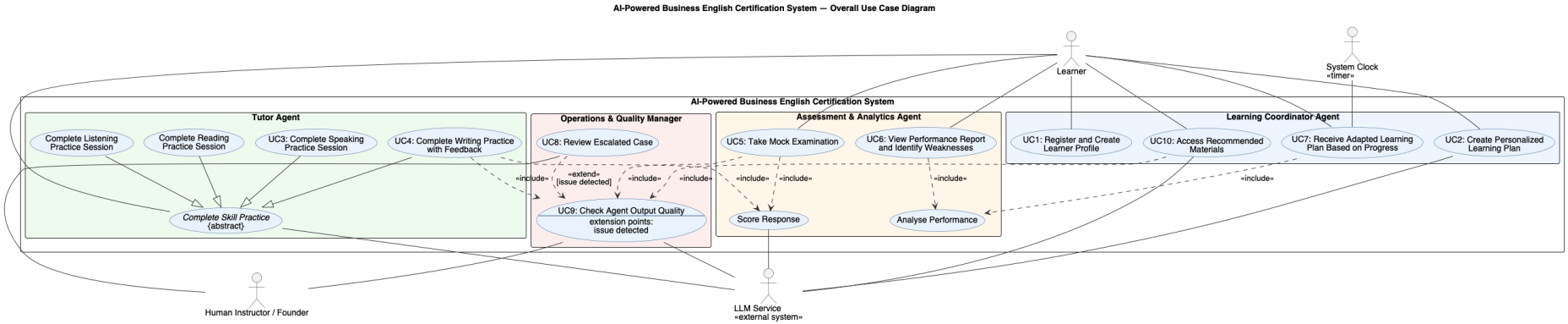


Figure 2: Overall use case diagram.

3.2 Use Case Specifications

Each member wrote at least one specification. The template combines the header from Tutorial 3 (name, goal, level, precondition, success and failure conditions, trigger) with the event flows from Tutorial 4.

UC1: Register and Create Learner Profile *Author: Menglu Lin*

Attribute	**Details**
Name	Register and Create Learner Profile
Goal	Learner creates an account and provides initial profile information for personalisation
Level	User goal
Precondition	Learner is new to the system and not yet registered, or an existing learner is updating their profile
Success Condition	Learner profile is created or updated and stored; learner is confirmed by the system
Failure Condition	Registration incomplete (missing required fields); account creation fails
Trigger	Learner clicks "Register" or "Sign Up"
User	Learner
Basic Flow	1. The learner selects "Register". 2. The system displays the registration form and requests the required account and learner profile information. 3. The learner enters their account details, target exam, current proficiency level, target score, expected exam date, and available study time. 4. The learner submits the completed registration form. 5. The system checks that all required information is present and valid. 6. The system creates the learner account. 7. The Learning Coordinator Agent creates and stores the learner profile using the submitted information. 8. The system displays the completed learner profile to the learner for review. 9. The learner confirms that the profile information is correct. 10. The system marks the learner profile as confirmed and makes it available for personalised learning plan creation.

UC2: Create Personalized Learning Plan *Author: Menglu Lin*

Attribute	**Details**
Name	Create Personalized Learning Plan
Goal	AI Learning Coordinator generates a customised study plan aligned with the learner's goals
Level	User goal

Attribute	Details
Precondition	Learner profile complete (UC1 completed); learner has confirmed their profile information
Success Condition	Learning plan created and stored; plan displayed to the learner with its rationale; learner confirms acceptance of the plan
Failure Condition	Plan generation fails; insufficient data to create a plan; learner rejects the plan
Trigger	Learner starts a "Create Learning Plan" request after confirming their profile
User	Learner
Basic Flow	1. The learner selects "Create Learning Plan" after confirming their learner profile. 2. The system retrieves the learner's confirmed profile information. 3. The Learning Coordinator Agent reviews the learner's target exam, target score, current proficiency level, identified weaknesses, expected exam date, and available study time. 4. The Learning Coordinator Agent determines which skills and learning activities should be prioritised. 5. The Learning Coordinator Agent generates a personalised learning plan based on the learner's profile and priorities. 6. The system stores the generated learning plan. 7. The system displays the personalised learning plan together with the rationale for its main priorities and recommended activities. 8. The learner reviews the proposed learning plan. 9. The learner confirms acceptance of the learning plan. 10. The system marks the plan as active and makes it available for subsequent learning and practice activities.

UC3: Complete Speaking Practice Session *Author: Suhani Saha*

Attribute	Details
Name	Complete Speaking Practice Session
Goal	Learner practises speaking skills and receives immediate feedback on pronunciation, fluency, vocabulary, grammar and structure
Level	User goal
Precondition	Learner profile exists (UC1 completed); learning plan exists (UC2 completed); speaking is included in the learner's current priorities
Success Condition	Speaking practice is completed; learner performance is evaluated; verified feedback and estimated scores are provided; session data is recorded
Failure Condition	Audio recording fails; feedback cannot be generated; learner quits before submission; a technical issue prevents the session from completing

Attribute	Details
Trigger	Learner selects “Speaking Practice” from the dashboard
User	Learner
Basic Flow	1. The learner selects “Speaking Practice”. 2. The system retrieves the learner’s profile, target exam, target score and current speaking priorities. 3. The AI Tutor prepares an appropriate IELTS or TOEFL speaking task based on the learner’s current level and weaknesses. 4. The system presents the speaking prompt together with the required preparation and speaking time. 5. The learner prepares, records and submits the speaking response. 6. The AI Tutor analyses the response and generates feedback on pronunciation, fluency, vocabulary, grammar, hesitations and structure. 7. The Assessment & Analytics Agent evaluates the speaking performance and produces estimated criterion scores with confidence values. 8. The system includes UC9: Check Agent Output Quality to verify that the Tutor feedback and Assessment scores are consistent and sufficiently confident. 9. After UC9 verifies the result, the system displays the feedback and estimated scores to the learner. 10. The system stores the speaking-session result and updates the learner’s performance history.
Alternate Flows	**3a. The learner has no recent speaking weakness recorded:** generate a speaking task based on the learner’s current level, target exam and learning plan. **5a. Learner exits before submitting the recording:** save the available session progress where possible and record the practice session as incomplete. **8a. UC9 holds the result:** do not release the feedback or estimated scores as final; inform the learner that the result is being double-checked while UC9 follows its escalation process. **9a. Learner chooses to repeat the task:** allow the learner to re-record the same speaking task and compare the new performance with the previous attempt.

Attribute	Details
Exception Flows	**4a. Speaking prompt cannot be loaded:** retry once; if the task still cannot be provided, stop the session and notify the learner that speaking practice is temporarily unavailable. **5b. Audio recording or upload fails:** preserve any available recording data, allow the learner to retry the recording or upload, and do not continue to evaluation until a valid response is received. **6a. Tutor feedback generation fails:** retry feedback generation once; if it still fails, keep the submitted response and notify the learner that feedback is temporarily unavailable. **7a. Speaking assessment cannot be completed:** hold the result and retry scoring once; if scoring still fails, do not release an estimated score and record the failure for review.

UC4: Complete Writing Practice with Feedback *Author: Suhani Saha*

Attribute	Details
Name	Complete Writing Practice with Feedback
Goal	Learner completes a writing exercise and receives detailed, trustworthy feedback on grammar, vocabulary, structure and task response
Level	User goal
Precondition	Learner profile exists (UC1 completed); learning plan exists (UC2 completed); writing is included in the learner's current priorities
Success Condition	Writing exercise is completed and submitted; detailed feedback and estimated criterion scores are verified and provided; learner receives relevant improvement recommendations; submission data is recorded
Failure Condition	Writing submission fails; feedback cannot be generated; quality verification fails; learner abandons the exercise without submitting
Trigger	Learner selects "Writing Practice" from the dashboard
User	Learner

Attribute	Details
Basic Flow	1. The learner selects "Writing Practice". 2. The system retrieves the learner's profile, target exam, target score and current writing priorities. 3. The AI Tutor prepares an IELTS or TOEFL writing task appropriate to the learner's current level and weaknesses. 4. The system presents the writing prompt together with the timer and word counter. 5. The learner writes and submits the completed response. 6. The AI Tutor analyses the response and generates detailed feedback on grammar, vocabulary, task response, coherence, structure and paragraph linking. 7. The Assessment & Analytics Agent evaluates the response against the relevant IELTS or TOEFL rubric and produces criterion scores with confidence values. 8. The system includes UC9: Check Agent Output Quality to verify that the Tutor feedback and Assessment scores are consistent and sufficiently confident. 9. After UC9 verifies the result, the system displays the detailed feedback, estimated scores and improvement recommendations to the learner. 10. The system stores the writing result and updates the learner's performance history.
Alternate Flows	**3a. The learner has no recent writing weakness recorded:** generate a writing task based on the learner's current level, target exam and learning plan. **5a. Learner submits an incomplete response:** inform the learner that the response may not receive a reliable full assessment and allow the learner either to continue editing or submit it as incomplete practice. **8a. UC9 holds the result:** do not release the feedback or estimated scores as final; inform the learner that the result is being double-checked while UC9 follows its escalation process. **9a. Learner chooses to revise the same response:** allow the learner to edit and resubmit the essay; compare the new result with the previous attempt and show which problems have improved or remain.

Attribute	Details
Exception Flows	**4a. Writing prompt cannot be loaded:** retry once; if the task still cannot be provided, stop the session and notify the learner that writing practice is temporarily unavailable. **5b. Submission fails:** preserve the learner's draft where possible, allow the learner to retry submission and do not continue to evaluation until the response is successfully received. **6a. Tutor feedback generation fails:** retry feedback generation once; if it still fails, keep the submitted essay and notify the learner that feedback is temporarily unavailable. **7a. Writing assessment cannot be completed:** hold the result and retry scoring once; if scoring still fails, do not release estimated bands and record the failure for review. **8b. UC9 cannot complete the quality check:** keep the result on hold and follow the exception or escalation process defined in UC9 rather than releasing unverified feedback to the learner.

UC5: Take Mock Examination

Author: Yuxuan Tang

Attribute	Details
Name	Take Mock Examination
Goal	Learner completes a full-length mock IELTS/TOEFL exam; the system evaluates performance and predicts a band score
Level	User goal
Precondition	Learner profile exists; learning plan exists; learner has completed sufficient preparation
Success Condition	Mock exam completed across all four skills; scored and evaluated; comprehensive report generated; predicted band score provided
Failure Condition	Exam not completed; technical failure during the exam; learner exits before completion
Trigger	Learner selects "Take Mock Examination" and confirms they are ready to begin
User	Learner

Attribute	Details
Basic Flow	1. Learner starts the mock examination. 2. The system checks that the learner meets the required preparation conditions. 3. The AI Assessment & Analytics Agent prepares the IELTS/TOEFL mock examination. 4. Learner completes the Listening, Reading, Writing, and Speaking sections. 5. The AI Assessment & Analytics Agent scores the completed sections. 6. The agent analyses the learner's performance and identifies key strengths and weaknesses. 7. The agent predicts the learner's IELTS/TOEFL score and exam readiness. 8. The AI Quality & Operations Agent checks the assessment result for major inconsistencies or low-confidence outcomes. 9. The system stores the assessment result and makes it available to the learner. 10. Learner views the result and predicted score.
Alternate Flows	**2a. Preparation requirements are not met:** The system informs the learner that the mock examination cannot start yet. The learner returns to further preparation. **4a. Learner does not complete the examination:** The system records the examination as incomplete. No final mock-exam result is provided. **4b. A technical problem occurs during the examination:** The system saves the available examination progress. The learner is informed of the problem and may retry or resume where possible. **8a. The assessment result has low confidence or appears inconsistent:** The result is flagged by the AI Quality & Operations Agent. The case is sent to the Human Founder/Teacher for review before the result is confirmed.

Attribute	Details
Exception Flows	**3a. Mock examination generation fails:** retry the examination generation once; if it still fails, stop the examination setup and notify the learner that the mock exam is temporarily unavailable. **5a. One or more sections cannot be scored:** hold the final result, preserve the completed responses and retry the scoring process. If scoring still fails, flag the affected section for Human Founder/Teacher review. **8b. Quality verification cannot be completed:** do not release the assessment result as final; mark it as pending verification and send it for Human Founder/Teacher review. **9a. Assessment result cannot be stored:** keep the generated result temporarily, log the storage failure and retry saving before making the result available to the learner.

UC6: View Performance Report and Identify Weaknesses*Author: Yuxuan Tang*

Attribute	Details
Name	View Performance Report and Identify Weaknesses
Goal	Learner accesses comprehensive performance data to understand their proficiency level and the areas needing improvement
Level	User goal
Precondition	Learner profile exists; learner has completed at least one practice activity or assessment
Success Condition	Performance report generated and displayed; weakness analysis provided; recommendations shown
Failure Condition	Insufficient assessment data; report generation fails
Trigger	Learner selects "Performance Report" or "Progress Dashboard" from the menu
User	Learner

Attribute	Details
Basic Flow	1. Learner opens the Performance Report or Progress Dashboard. 2. The system retrieves the learner's available practice and assessment results. 3. The AI Assessment & Analytics Agent analyses performance across Listening, Reading, Writing, and Speaking. 4. The agent identifies the learner's main strengths, weaknesses, and repeated problem areas. 5. The agent reviews the learner's progress using previous results. 6. The system generates relevant recommendations for improvement. 7. The AI Quality & Operations Agent checks the report for inconsistent or low-confidence information. 8. The system displays the completed performance report. 9. Learner reviews the report and understands the areas that need more improvement.
Alternate Flows	**2a. There is not enough assessment data:** The system informs the learner that more practice or assessment is needed before a complete report can be generated. **2b. Assessment data cannot be retrieved:** The system informs the learner that the report is temporarily unavailable. The learner can try again later. **7a. Conflicting or low-confidence information is found:** The affected result is flagged for review. If necessary, the issue is escalated to the Human Founder/Teacher. **8a. The report cannot be displayed:** The learner is informed that the report is currently unavailable. The system keeps the existing assessment data for a later attempt.
Exception Flows	**3a. Performance analysis fails:** retry the analysis once; if it still fails, stop report generation and inform the learner that the report cannot currently be completed. **6a. Recommendation generation fails:** generate the report using the available verified performance analysis, omit the unavailable recommendations and record the failure for retry. **7b. Quality verification cannot be completed:** do not present the affected analysis as verified; mark the report as pending review and escalate the relevant result to the Human Founder/Teacher. **8b. Report data becomes unavailable before display:** keep the underlying assessment records unchanged, log the failure and allow the learner to request the report again later.

UC7: Receive Adapted Learning Plan Based on Progress *Author: Mohamed Azim*

Attribute	Details
Name	Receive Adapted Learning Plan Based on Progress
Goal	Revise the learner's learning plan based on recent assessment data and adjust study priorities to reflect their current progress
Level	User goal
Precondition	An initial learning plan exists (UC2 completed); the learner has completed multiple assessments; and a weekly review, manual request or major assessment has triggered adaptation
Success Condition	An adapted learning plan is generated with a clear rationale; relevant resources are recommended inline; the learner confirms the new plan; and all relevant agents are notified of the changes
Failure Condition	There is insufficient data to justify an adaptation, the adaptation fails, or the learner rejects the adapted plan; the existing plan remains active
Trigger	Automatic weekly review OR learner requests an adaptation OR a major assessment is completed (UC5)
User	Primary: Learner. Secondary: LLM Service, Learning Coordinator
Basic Flow	1. The system retrieves the learner's current learning plan and recent assessment results. 2. The Learning Coordinator compares the learner's recent progress with the existing study priorities. 3. The system identifies areas where the learner's priorities should change. 4. The LLM Service proposes an adapted learning plan and provides a rationale for each major change. 5. The Learning Coordinator selects relevant study resources to support the new priorities. 6. The system presents the adapted plan, including the new priorities, rationale and recommended resources, to the learner. 7. The learner confirms the adapted plan. 8. The system activates the new plan and notifies the relevant agents of the updated priorities.
Alternate Flows	**3a. Insufficient evidence:** keep the existing learning plan and tell the learner that more assessment data is needed. **6a. Learner requests changes:** allow the learner to review the proposed priorities before confirming the plan. **7a. Learner rejects the adapted plan:** keep the existing plan active and record the rejection for future adaptation decisions.
Exception Flows	**4a. LLM Service does not respond:** retry once; if the adaptation cannot be generated, keep the existing plan and notify the learner. **5a. No suitable resources are found:** keep the new priorities but show the affected area without an inline recommendation until suitable material becomes available. **8a. Agent notification fails:** keep the confirmed plan active, log the failure and retry the notification.

UC8: Review Escalated Case *Author: Mohamed Azim*

Attribute	Details
Name	Human Instructor Reviews Escalated Case
Goal	Provide human expert judgment on complex, low-confidence or conflicting cases that cannot be safely resolved by the AI agents alone
Level	Subfunction
Precondition	A case has been flagged by the Quality & Operations Agent; an escalation notification has been sent to the instructor; and the case contains the available context and supporting evidence
Success Condition	The instructor reviews the case, makes a decision (approve, override or request coaching), the decision is documented, and the relevant agents are notified
Failure Condition	There is insufficient context, the instructor is unavailable, or a decision cannot be made; the case remains open and is not silently resolved
Trigger	AI Operations & Quality Manager flags a case and escalates it to the human instructor
User	Primary: Human Instructor/Founder. Secondary: LLM Service, AI Operations & Quality Manager
Basic Flow	1. The Quality & Operations Agent creates an escalation containing the reason for escalation and the available case context. 2. The system notifies the human instructor that a case requires review. 3. The instructor opens the case and reviews the learner's relevant assessment results, AI feedback, confidence values and previous decisions. 4. The system highlights any conflicting results or low-confidence decisions that caused the escalation. 5. The instructor reviews the evidence and decides whether to approve the AI result, override it, or request additional coaching/information. 6. The system records the instructor's decision and reasoning in the case history. 7. The system notifies the relevant agents of the instructor-approved outcome. 8. The case is marked resolved and the decision is available for future quality checks.
Alternate Flows	**5a. Instructor approves the AI result:** record the approval and release or continue processing the result according to the calling use case. **5b. Instructor overrides the AI result:** record the new decision and use the instructor-approved result instead of the original AI result. **5c. Instructor requests coaching or more information:** keep the case open, request the additional information and return the case for further review.

Attribute	Details
Exception Flows	**3a. Insufficient context:** keep the case open and request the missing learner data or supporting evidence before a decision can be made. **2a. Instructor unavailable:** keep the case in the escalation queue and notify an authorised instructor when available. **7a. Agent notification fails:** record the instructor's decision, keep the case resolved locally and retry notifying the relevant agents.

UC9: Check Agent Output Quality *Author: Naratorn Pisedtasalasai*

Attribute	Details
Name	Check Agent Output Quality
Goal	Ensure no AI-generated result (feedback, score or recommendation) reaches the learner unless the agents agree with each other and the AI is confident in it
Level	Subfunction (included by UC4, UC5 and UC10)
Precondition	An agent has produced a result with a confidence value, e.g. Tutor feedback plus Assessment band scores; a confidence threshold has been set by the instructor (default 0.7)
Success Condition	Result passes the confidence and consistency checks, is marked "Verified", released to the learner and logged
Failure Condition	Result is held and not released; the learner sees "Your result is being double-checked"; the case is escalated to the instructor (UC8) and logged
Trigger	UC4, UC5 or UC10 submits an AI-generated result for release to the learner
User	Primary: none (started by the calling use case). Secondary: LLM Service, Human Instructor (via UC8)
Basic Flow	1. The calling use case sends the result to the Operations & Quality Manager. 2. The system checks that every part of the result has a confidence value. 3. The system compares each confidence value with the threshold. All pass. 4. The system sends the Tutor's feedback and the Assessment scores to the LLM Service, asking whether they contradict each other. 5. The LLM Service returns a structured verdict (consistent / conflicts / reason). No conflicts are found. 6. The system marks the result "Verified" and releases it to the learner. 7. The system logs the check and updates the issue statistics.

Attribute	Details
Alternate Flows	**3a. Confidence below threshold:** hold the result, notify the learner, escalate to UC8. **5a. Contradiction found** (e.g. feedback praises vocabulary but Lexical Resource is 5.0): hold the result, attach the conflict, escalate to UC8. **7a. Same issue for 5+ learners on one topic in 7 days:** notify the instructor of a possible curriculum gap.
Exception Flows	**2a. Result is incomplete:** ask the agent to regenerate it once, otherwise hold and escalate. **4a. LLM Service does not respond:** retry once, then fall back to rule-based checks. If those cannot decide, hold and escalate.

UC10: Access Recommended Materials *Author: Naratorn Pisedtasalasai*

Attribute	Details
Name	Access Recommended Materials
Goal	Learner gets study materials that target their own weaknesses, with a clear reason for each recommendation
Level	User goal
Precondition	Learner is logged in; learner profile and learning plan exist (UC1 and UC2 completed)
Success Condition	Learner sees a ranked list of materials, each linked to a specific weakness, and opens at least one; access is recorded and sent to the Learning Coordinator
Failure Condition	No personalised list can be produced; the learner is shown general materials for their level and told why
Trigger	Learner selects "Recommended for Me" on the dashboard or in their performance report (UC6)
User	Primary: Learner. Secondary: LLM Service
Basic Flow	1. The learner selects "Recommended for Me". 2. The system retrieves the learner's weaknesses (Assessment & Analytics Agent) and plan priorities (Learning Coordinator). 3. The system searches the material library for matching items. 4. The system asks the LLM Service to rank the items and write a one-line reason for each. 5. The LLM Service returns the ranked list with reasons and confidence values. 6. The system includes UC9 to check that the recommendations match the learner's assessment results. 7. The system displays the top 5 materials with reasons. 8. The learner opens a material, and the system records it and informs the Learning Coordinator.

Attribute	Details
Alternate Flows	**2a. New learner with no assessment data:** show starter materials for their level and target exam, labelled "General". **3a. No library item matches a weakness:** the LLM generates a short targeted exercise, labelled "AI-generated", which is checked by UC9. **6a. UC9 holds the recommendations:** show the last verified list while the instructor reviews. **8a. Learner marks an item "Not relevant":** remove it and use the feedback in future rankings.
Exception Flows	**4a. LLM Service unavailable:** rank by weakness tags only and show the list without reasons. **8b. Material fails to load:** show an error and offer the next item.

4 Architecture Analysis and Design

4.1 Viewpoint Analysis

The architecture is described using Kruchten's 4+1 view model [3], in the sense of ISO/IEC/IEEE 42010 [4], where each viewpoint addresses the concerns of particular stakeholders.

Viewpoint	Concern / stakeholders	Models in this report
Use case (+1)	What the system must do for learners and the instructor	Feature diagram, use case diagram and specifications
Logical	Functionality: classes and their responsibilities (developers)	Class diagram, object diagram, collaboration diagram, sequence diagrams
Process	Runtime behaviour, concurrency and coordination of agents	Activity diagrams (with forks/joins), state machine diagrams
Development	Organisation of software modules	Packages in the class diagram (Agents, LLM Integration, Quality Control, etc.)
Physical	Deployment on servers and the external LLM API	Deployment is outside the scope of Stage 1; the LLM is modelled as an external system

Table 19: 4+1 viewpoints and the models that address them.

4.2 Architectural Style

The system uses a **layered architecture with centralised quality control**:

1. **Presentation layer:** learner and instructor interfaces.
2. **Agent layer:** the four agents, specialised from the abstract *Agent* class. Following the principle from Tutorial 4 of *separating control from function and centralising control*, the Operations & Quality Manager is a single control component that monitors the functional agents (Coordinator, Tutor, Assessment) and decides whether each result is released or held.
3. **Service layer:** the *LLMService* interface (implemented by *LLMProviderAdapter*), the *MaterialLibrary* and the *AuditLog*.
4. **Data layer:** learner profiles, learning plans, submissions, result packages and escalation cases.

4.3 Design Rationale and Discarded Alternatives

Decision	**Discarded alternative**	**Rationale**
All agents use one *LLMService* interface	Each agent calls an LLM API directly	The provider can be changed (or mocked for testing) without changing any agent; timeouts and retries are handled in one place.
Results are held until checked (UC9)	Release immediately and correct later	A wrong score damages learner trust and could mislead exam preparation; holding is safer.
Only low-confidence or contradictory results go to the instructor	The instructor reviews every result	One person cannot review every result; targeted escalation keeps human oversight scalable.
LLM returns structured output (JSON with confidence values)	Free-text LLM responses	Structured output can be checked automatically and compared between agents.
Rule-based fallback when the LLM fails	Block the learner until the LLM responds	The system stays usable during outages without releasing unchecked results.

Decision	Discarded alternative	Rationale
Four specialised agents	One general-purpose agent	Separate responsibilities make each agent simpler to prompt, test and monitor, and allow one agent to check another.

Table 20: Design decisions and discarded alternatives.

4.4 Support Components and Relationships for Future Development

- **LLMProviderAdapter:** wraps the external LLM API, handles timeouts, retries and structured output, and removes personal details before requests are sent.
- **MaterialLibrary:** stores study materials and supports search by weakness, level and exam type.
- **AuditLog:** records every quality check, escalation and instructor decision for auditing and for change assessment (Section 9).
- **LearnerProfile:** the shared state that all agents read and update.

The relationships between components support future development:

- A **new agent** (e.g. a pronunciation specialist) is added by specialising *Agent*; it is automatically monitored by the Operations & Quality Manager through the *monitors* association.
- A **new LLM provider** is added by implementing *LLMService*, with no change to the agents.
- A **new exam** (e.g. PTE) is added as a value of the *ExamType* enumeration and a new alternative in the feature diagram.
- A **new kind of AI output** is protected by including UC9 in its use case.

5 Structure Models

5.1 Elementary Structure: Class Diagram

Figure 3 shows the design-time class diagram [5], organised into packages. The main relationships are:

- **Generalisation:** the abstract class *Agent* is specialised into *LearningCoordinatorAgent*, *TutorAgent*, *AssessmentAgent* and *OpsQualityManager*.

- **Interface:** *LLMService* is an interface realised by *LLMProviderAdapter*; every *Agent* uses exactly one *LLMService*.
- **Composition:** parts that cannot exist without their whole, for example *Learner* ♦ *LearnerProfile* ♦ *LearningPlan* ♦ *StudyTask* {ordered}, *ResultPackage* ♦ *Feedback* and *BandScore*, and *RecommendationList* ♦ 1..5 *RankedItem* {ordered}.
- **Aggregation:** *MaterialLibrary* ◇ *Material*, since materials exist independently of the library.
- **Association class:** *Review* holds the instructor's decision about an *EscalationCase*; it belongs to the relationship, not to either class.
- **Multiplicity** is given on every association end, and enumerations (e.g. *ResultStatus*, matching the states of the UC9 state machine) constrain attribute values.

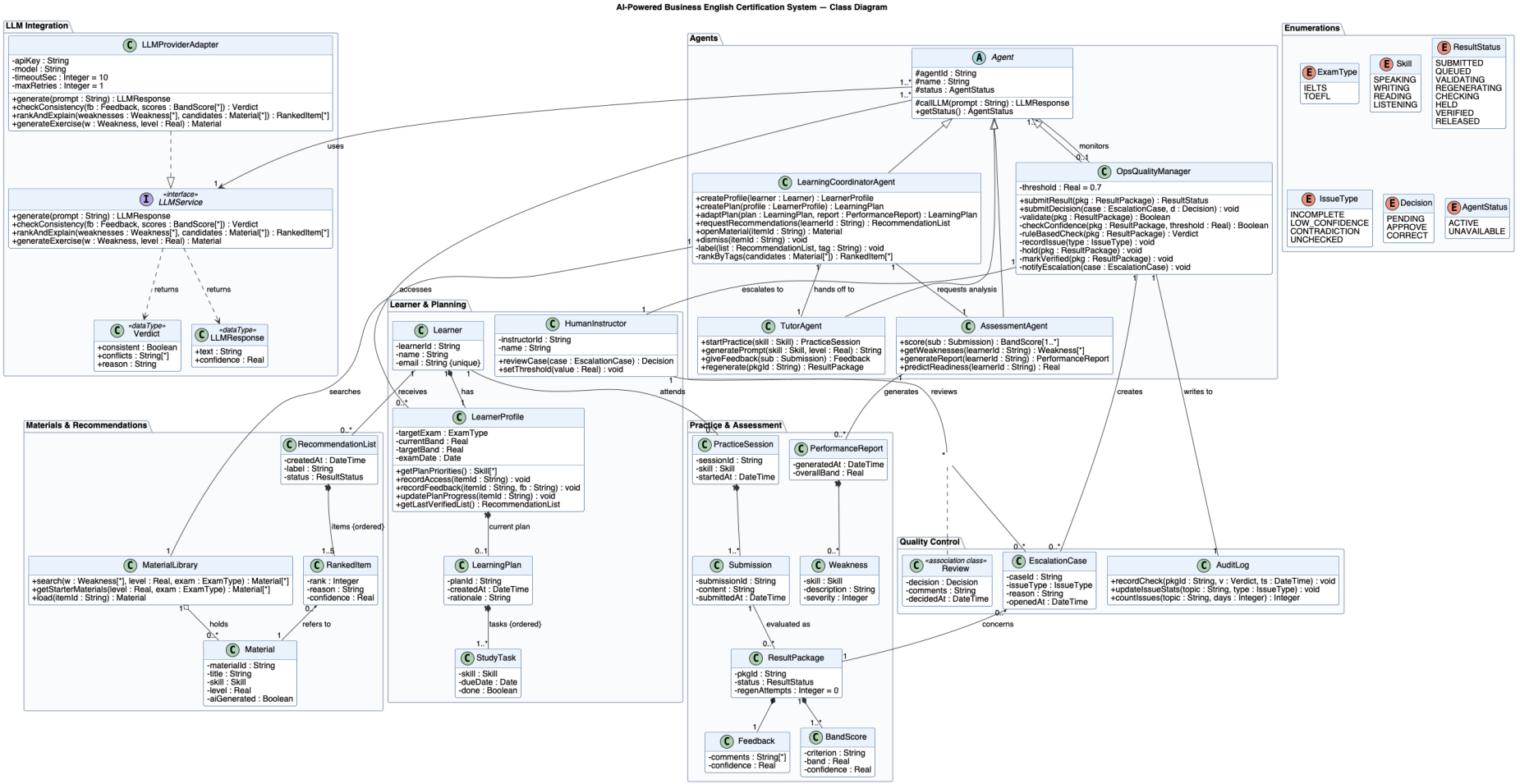


Figure 3: Class diagram.

5.2 Complex Structure: Object Diagram

Figure 4 is a runtime snapshot taken while UC9 is running: *learner Lili's diagnostic essay has been scored, a contradiction was found, and the result is HELD while an escalation case waits for the instructor.* The feedback praises Lili's vocabulary while the Lexical Resource band is 5.0, so *pkg1* is HELD and *case1* has issue type CONTRADICTION. All confidence values are at least 0.7, so the result was held because of the contradiction, not low confidence. One *llm1* object and one *profile1* object are shared by several agents and are drawn once. Every object is an instance of a class in Figure 3, every link is an instance of one of its associations, and all multiplicities are satisfied.

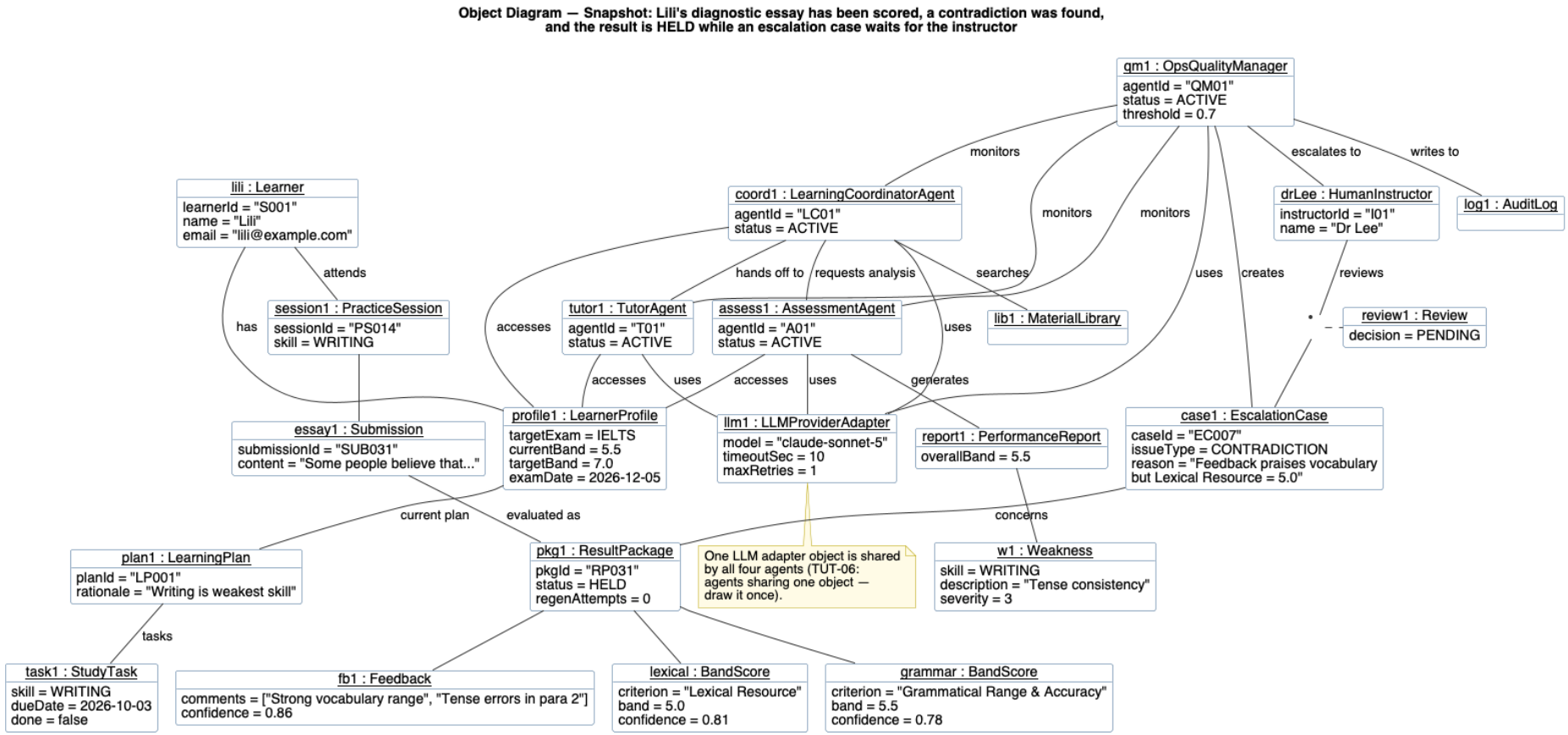


Figure 4: Object diagram: snapshot of a held result awaiting instructor review.

5.3 Complex Structure: Collaboration Diagram

Figure 5 abstracts the snapshot into the collaboration *QualityCheckedEssayFeedback*. It shows the roles that must cooperate to deliver quality-checked feedback, independent of which objects fill them: the four agents, the shared LLM service and learner profile, the audit log and the instructor. Each connector corresponds to an association in the class diagram, and the table maps each role to the object that plays it in Figure 4. The UC9 and UC10 sequence diagrams (Section 6) are interactions that take place within this collaboration.

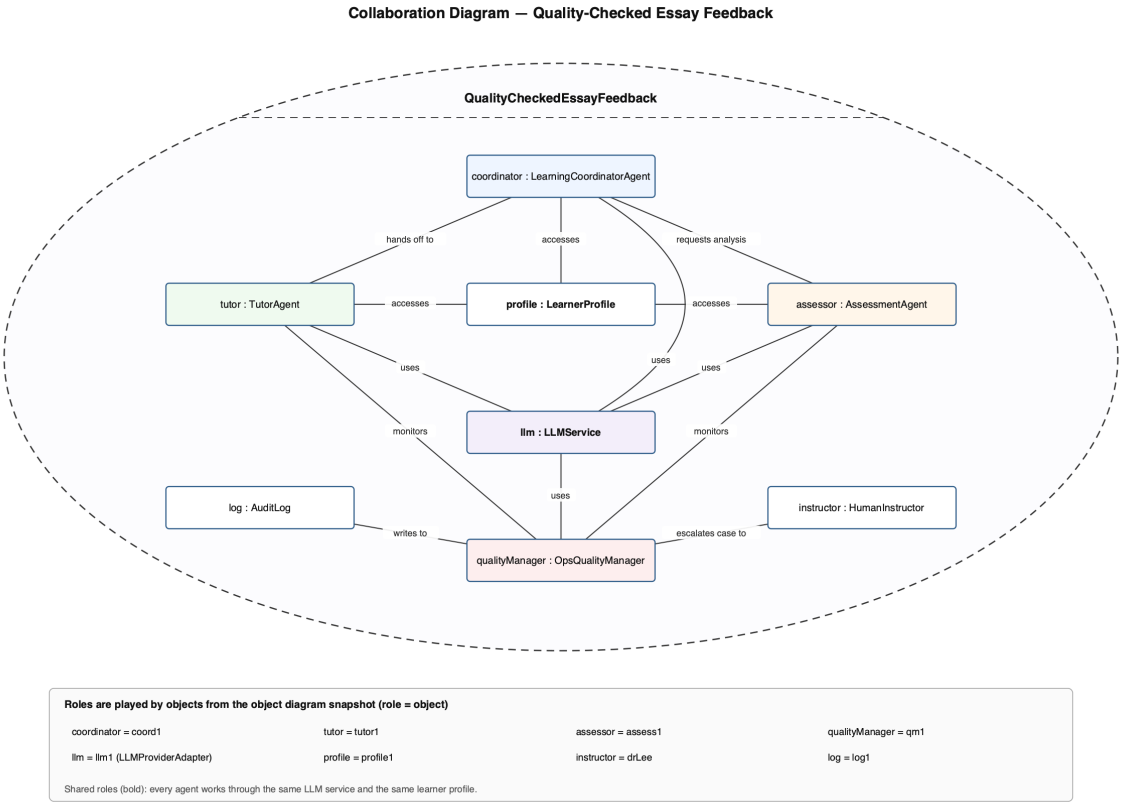


Figure 5: Collaboration diagram: QualityCheckedEssayFeedback.

6 Behaviour Models

Each member modelled the behaviour of their own use case with an activity diagram, an interaction (sequence) diagram and a state machine diagram.

6.1 UC9 and UC10 (Naratorn Pisedtasalasai)

6.1.1 UC9 Activity Diagram

Figure 6 has swimlanes for the Learner, the Tutor/Assessment Agent, the Operations & Quality Manager, the LLM Service and the Human Instructor. Each check (completeness, confidence, consistency) records an issue rather than jumping straight to "hold", and all checks lead to a single *Issue recorded?* decision. This gives one escalation path, so no result can skip the check. Fork/join bars show parallel work: notifying the learner while the instructor reviews, and releasing the result while logging the check.

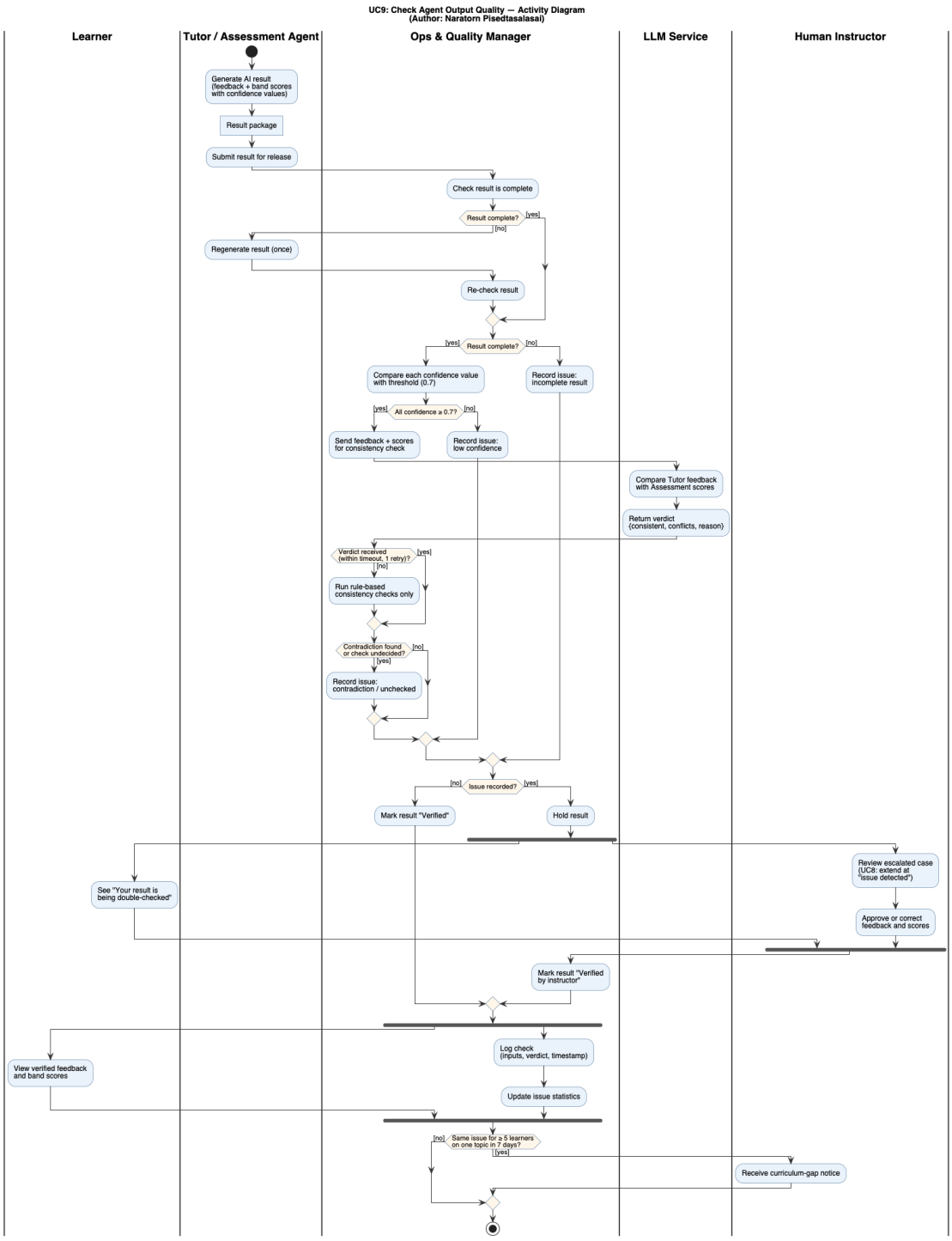


Figure 6: UC9 activity diagram.

6.1.2 UC9 Sequence Diagram

Figure 7 shows the messages between *tutor*, *qm*, *llm*, *log* and the actors. The LLM's role is the synchronous *checkConsistency()* call, which returns a structured verdict. Combined fragments show the logic: **opt** (regenerate an incomplete result), nested **alt** (confidence, and LLM timeout with rule-based fallback), **alt** (release vs escalate, with an *Escalation-*

Case created and a **ref** to UC8), **par** (logging) and **opt** (curriculum-gap notice).

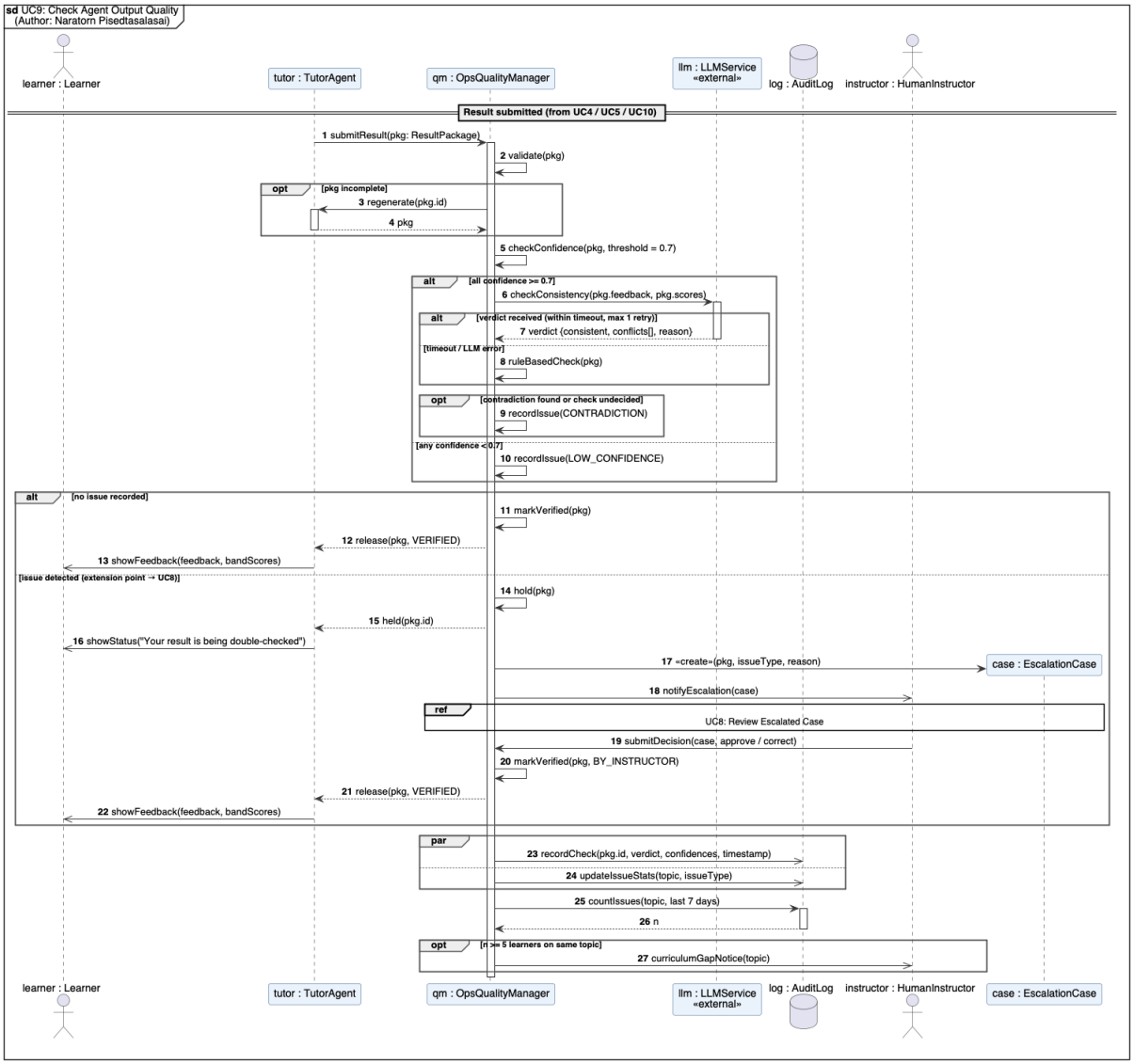


Figure 7: UC9 sequence diagram.

6.1.3 UC9 State Machine Diagram

Figure 8 follows a *ResultPackage* through its lifecycle: Submitted, Queued (if the quality manager is unavailable), Validating, Regenerating, Checking, Held, Verified and Released. *Checking* and *Held* are compound states. Transitions use the *trigger [guard] / effect* notation, and the diagram includes message events (*verdictReceived*), a change event (*when(QM available)*) and time events (*after(timeout)*, *after(24h)*).

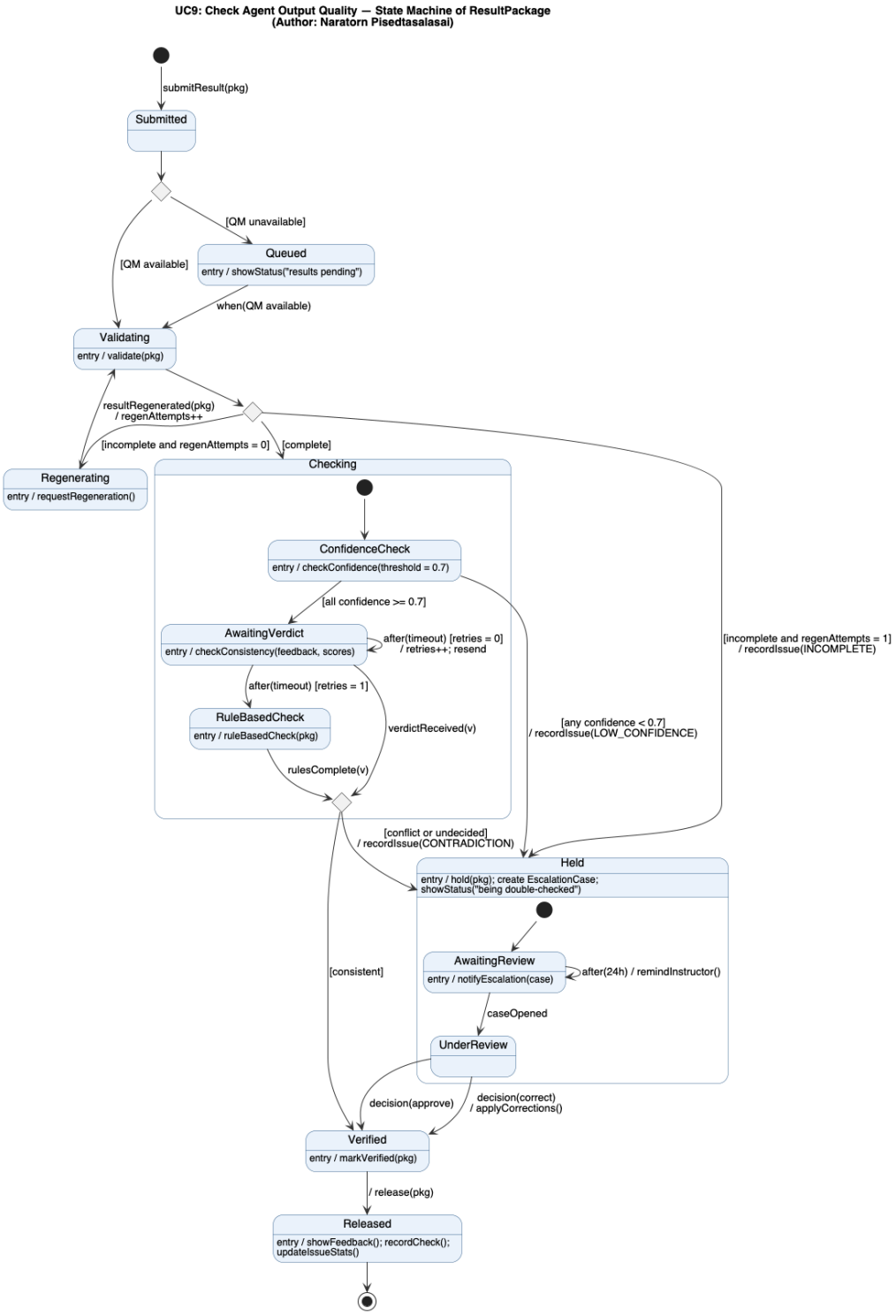


Figure 8: UC9 state machine diagram (ResultPackage).

6.1.4 UC10 Activity Diagram

Figure 9 shows the coordinator retrieving weaknesses and plan priorities in parallel, then searching the library, generating an exercise with the LLM when no item matches, and ranking with a fallback when the LLM times out. The list is checked by UC9 and the learner then browses in a loop.

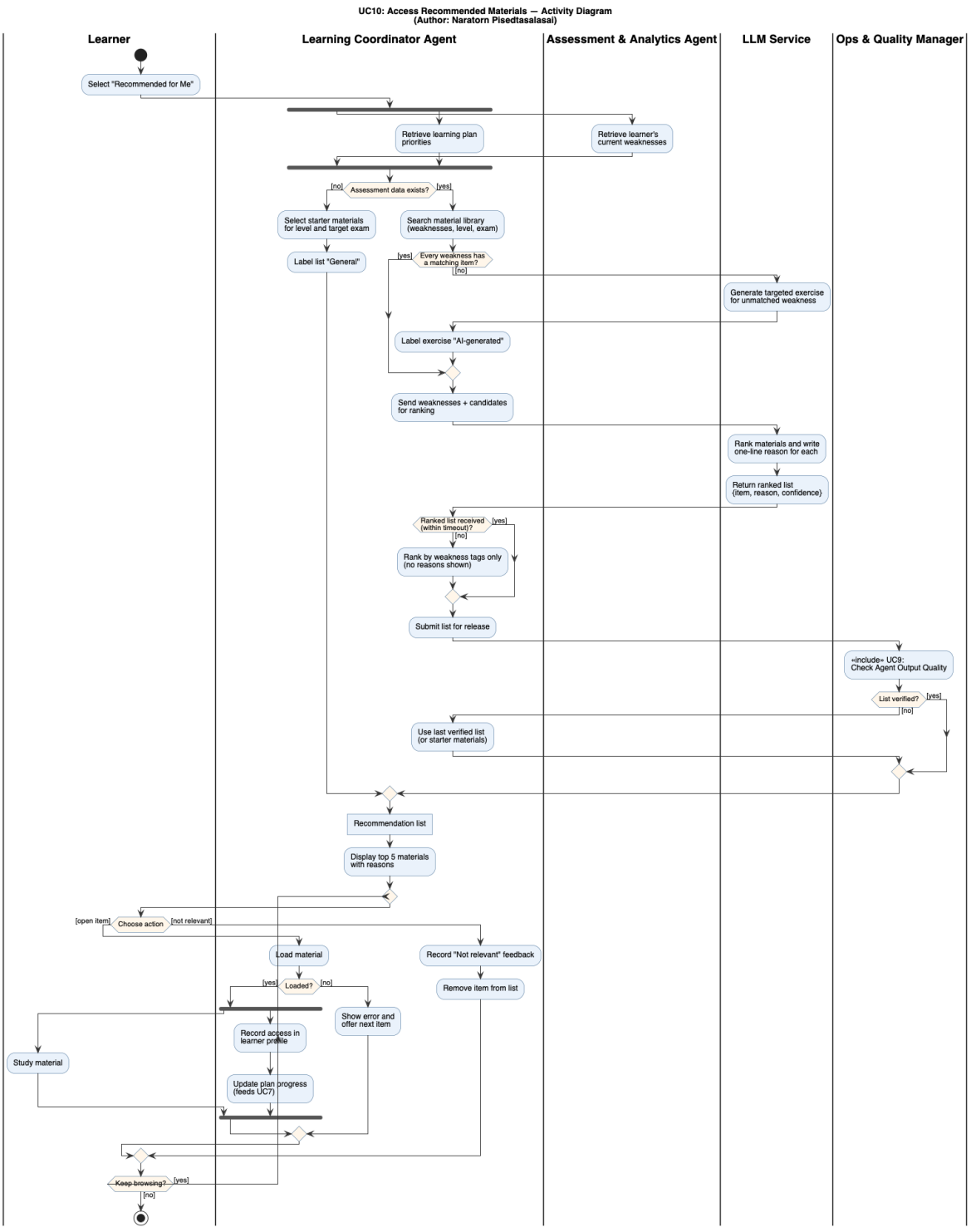


Figure 9: UC10 activity diagram.

6.1.5 UC10 Sequence Diagram

Figure 10 shows the same flow as messages. It uses **par**, **alt**, **opt**, **ref** (UC9) and **loop** fragments, and the LLM calls *generateExercise()* and *rankAndExplain()*.

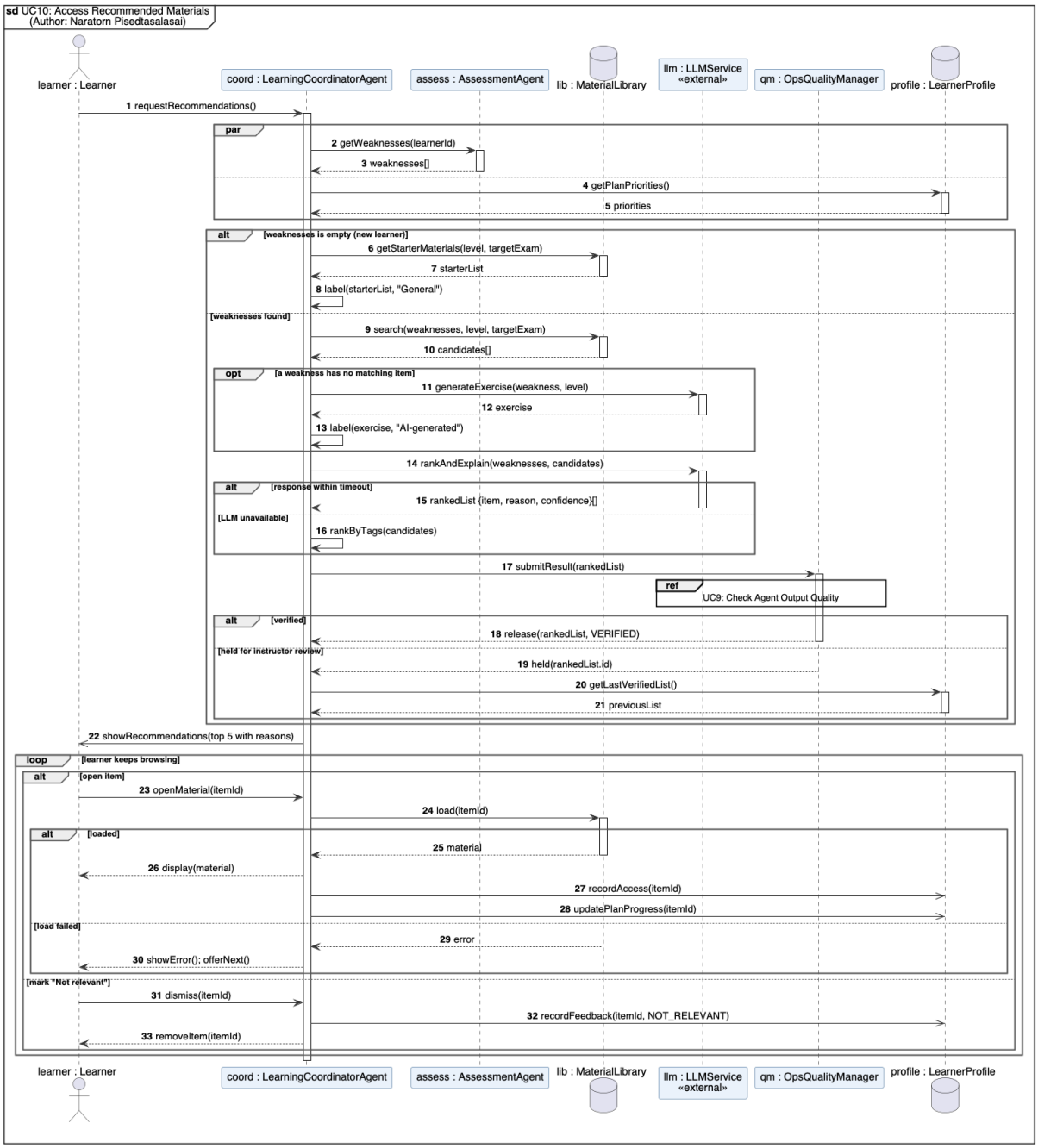


Figure 10: UC10 sequence diagram.

6.1.6 UC10 State Machine Diagram

Figure 11 follows a *RecommendationList* from Requested, through gathering candidates, generating exercises, ranking and the quality check, to Displayed. Displayed is a compound state with Browsing and ItemOpen substates, and it has a self-transition for dismissing an item.

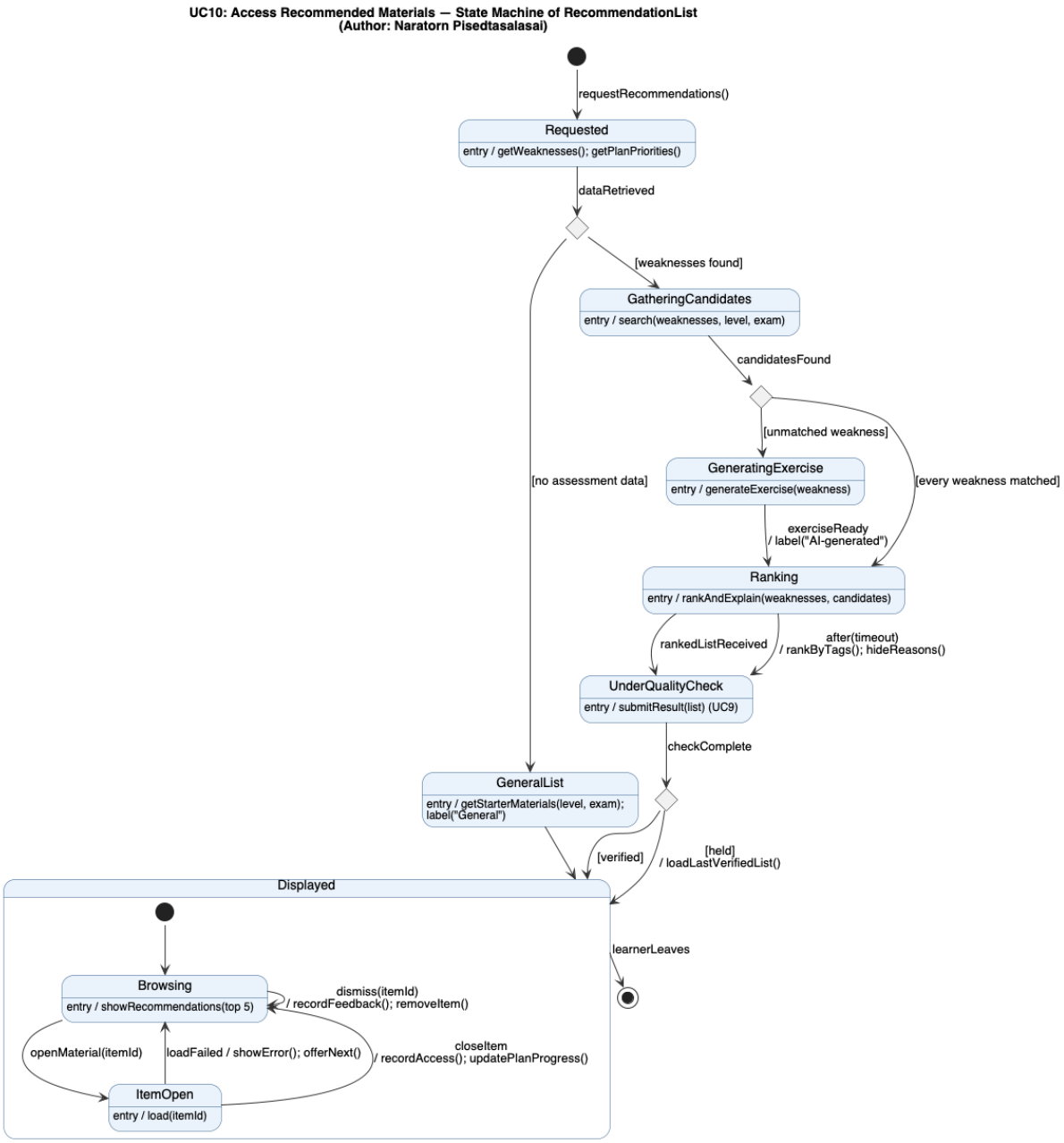


Figure 11: UC10 state machine diagram (RecommendationList).

6.2 UC1 and UC2 (Menglu Lin)

6.2.1 UC1 Activity Diagram

Figure 12 shows the registration and learner-profile creation process. It includes validation of submitted information, correction of invalid entries, and confirmation of the completed profile.

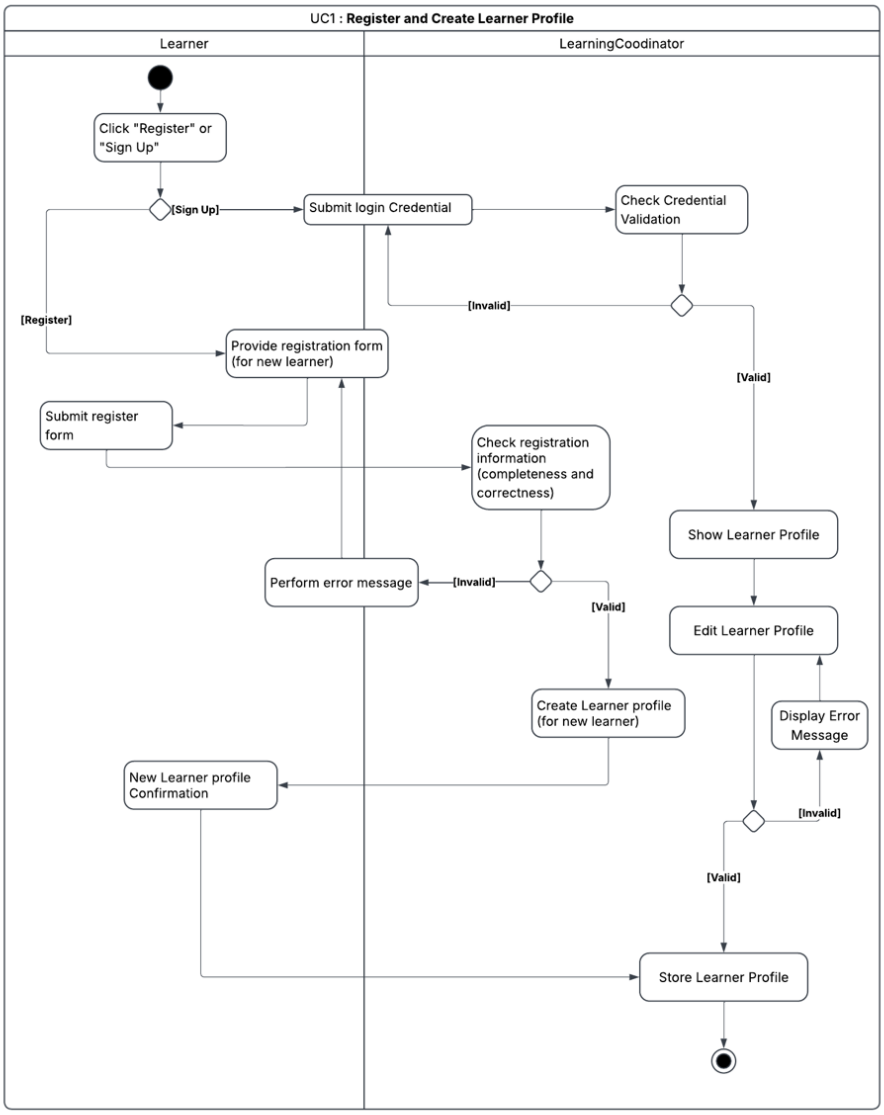


Figure 12: UC1 activity diagram.

6.2.2 UC1 Sequence Diagram

Figure 13 shows the interaction between the Learner, the Learning Coordinator Agent and profile storage. The **alt** and **loop** fragments represent validation outcomes and repeated correction of invalid data.

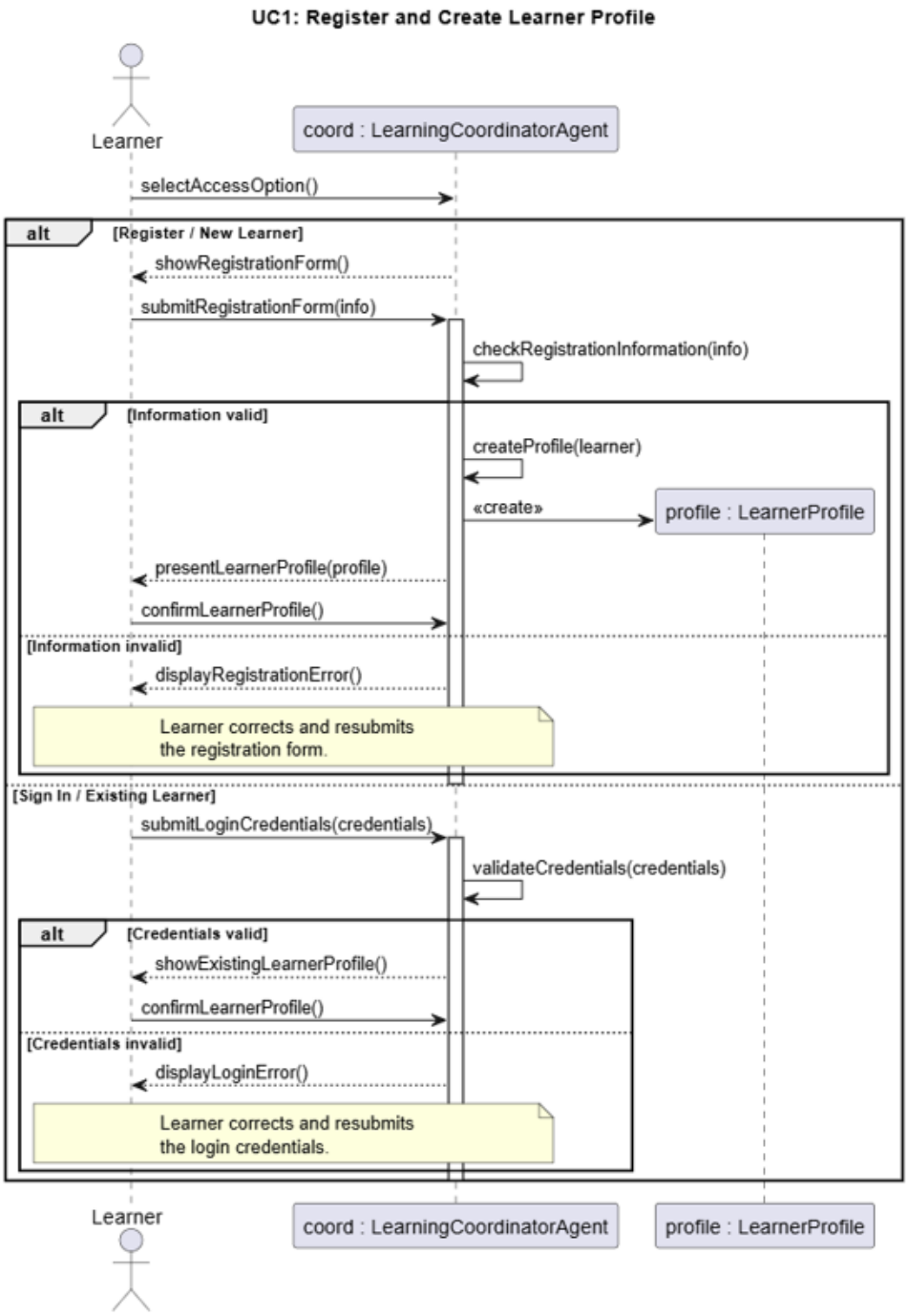


Figure 13: UC1 sequence diagram.

6.2.3 UC1 State Machine Diagram

Figure 14 shows the lifecycle of a *LearnerProfile*. The main states are *New*, *Validated*, *Confirmed* and *Updated*.

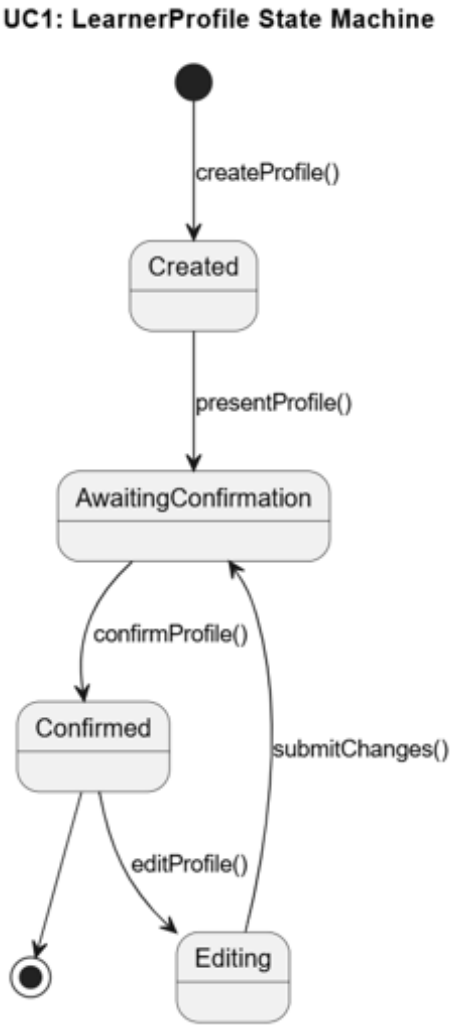


Figure 14: UC1 state machine diagram (LearnerProfile).

6.2.4 UC2 Activity Diagram

Figure 15 shows how a personalised learning plan is created from the confirmed learner profile. It includes profile analysis, plan generation, learner review and final plan acceptance.

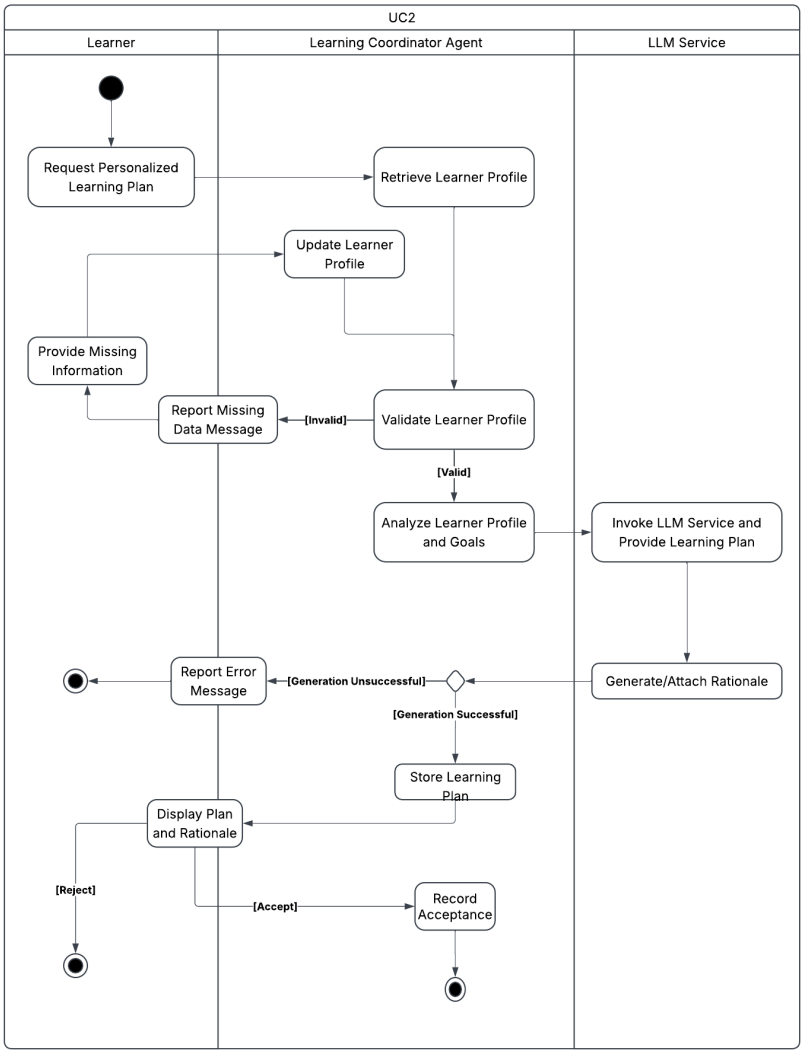


Figure 15: UC2 activity diagram.

6.2.5 UC2 Sequence Diagram

Figure 16 shows the messages used to generate and confirm a personalised learning plan. The **alt** and **loop** fragments represent insufficient profile information and repeated plan revision.

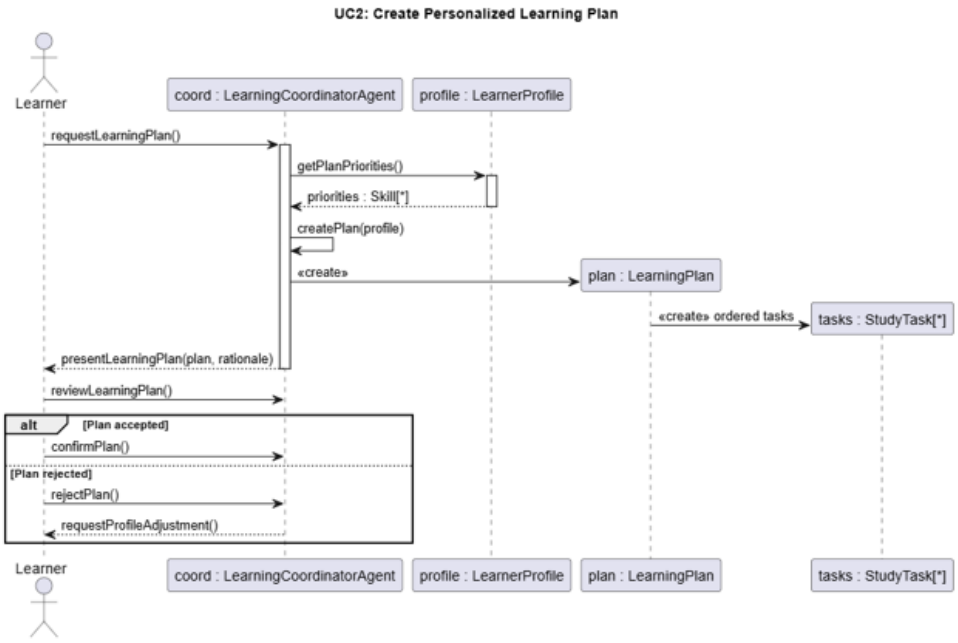


Figure 16: UC2 sequence diagram.

6.2.6 UC2 State Machine Diagram

Figure 17 shows the lifecycle of a *LearningPlan*. The plan moves from request and generation to proposal, revision and final activation.

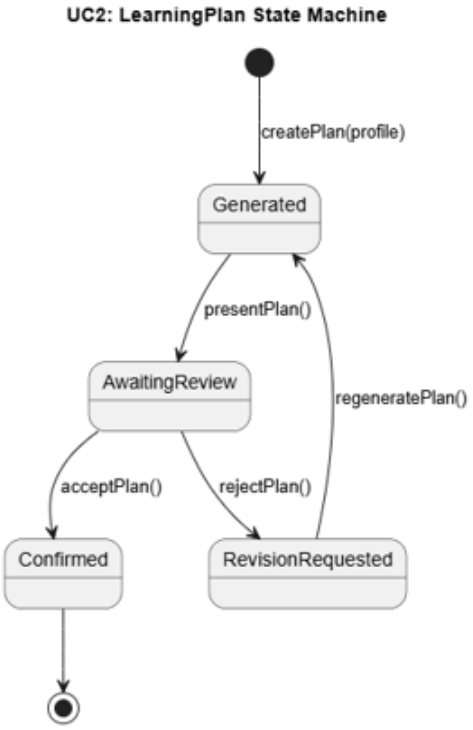


Figure 17: UC2 state machine diagram (LearningPlan).

6.3 UC5 and UC6 (Yuxuan Tang)

6.3.1 UC5 Activity Diagram

Figure 18 shows the activity flow for taking a mock examination across the *Learner*, *AI Assessment & Analytics Agent*, *AI Quality & Operations Agent*, and *Human Founder/Teacher* swimlanes. The learner first confirms readiness, after which the assessment agent checks the learner profile, learning plan, preparation level, and exam availability. If the requirements are satisfied, the learner completes the Listening, Reading, Writing, and Speaking sections and submits the responses. The assessment agent then verifies completion, scores the four sections, analyses performance, and predicts the IELTS/TOEFL score and exam readiness. The quality agent checks the result for inconsistency or low confidence; reliable results continue to report generation, while unreliable results are escalated to the human reviewer. The diagram also shows exception paths for incomplete sections, learner exit, technical failure, recovery, and resuming the examination from a saved section.

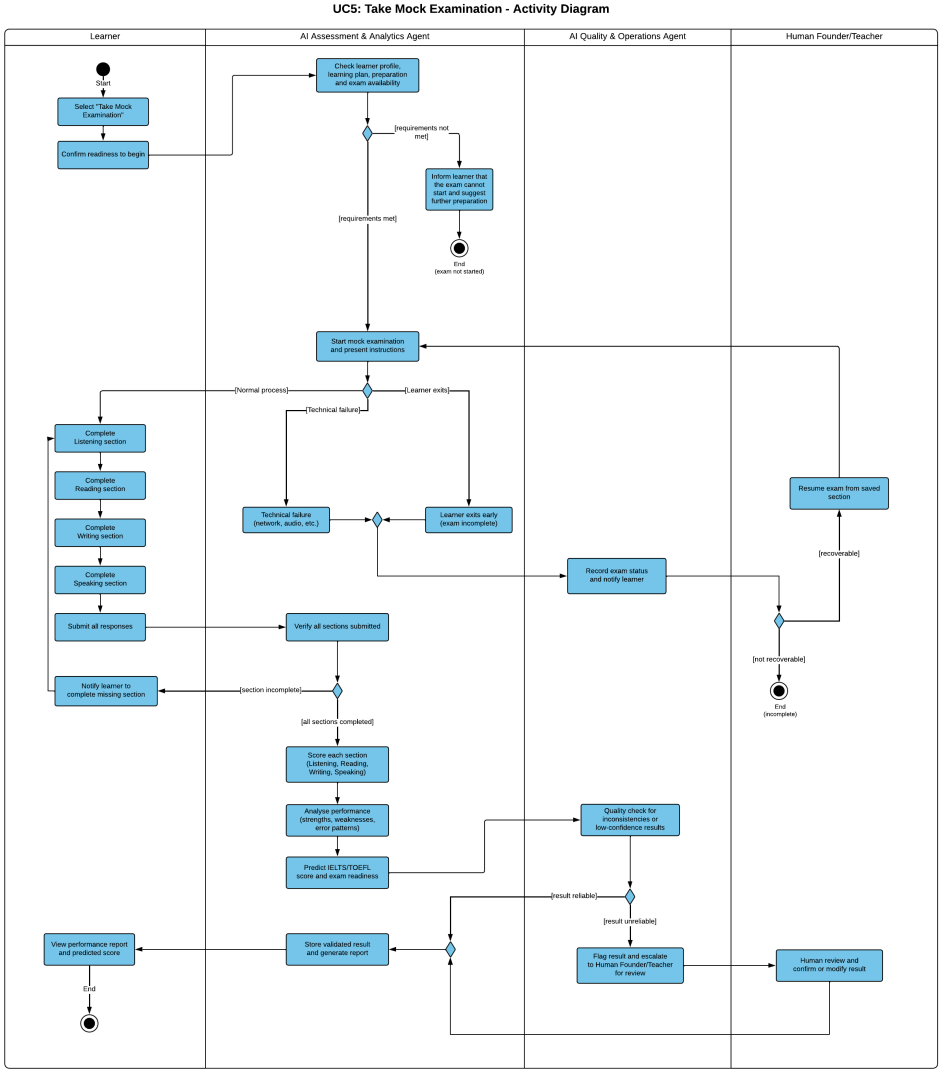


Figure 18: UC5 activity diagram.

6.3.2 UC5 Sequence Diagram

Figure 19 shows the interaction flow between the *Learner*, *AI Assessment & Analytics Agent*, *AI Quality & Operations Agent*, and *Human Founder/Teacher*. The first **alt** fragment separates the cases where the preparation requirements are satisfied or not satisfied. A **loop** fragment represents the repeated processing of the four examination sections, with a nested **alt** covering normal completion, early learner exit, and technical failure. Technical failure contains another **alt** for recoverable and non-recoverable cases. After all sections are completed, the assessment agent verifies completion, scores the exam, analyses strengths and weaknesses, and predicts score and readiness. A final **alt** distinguishes a reliable assessment result from a low-confidence or inconsistent result, in which case the quality agent escalates the result to the human reviewer before the validated report is returned to the learner.

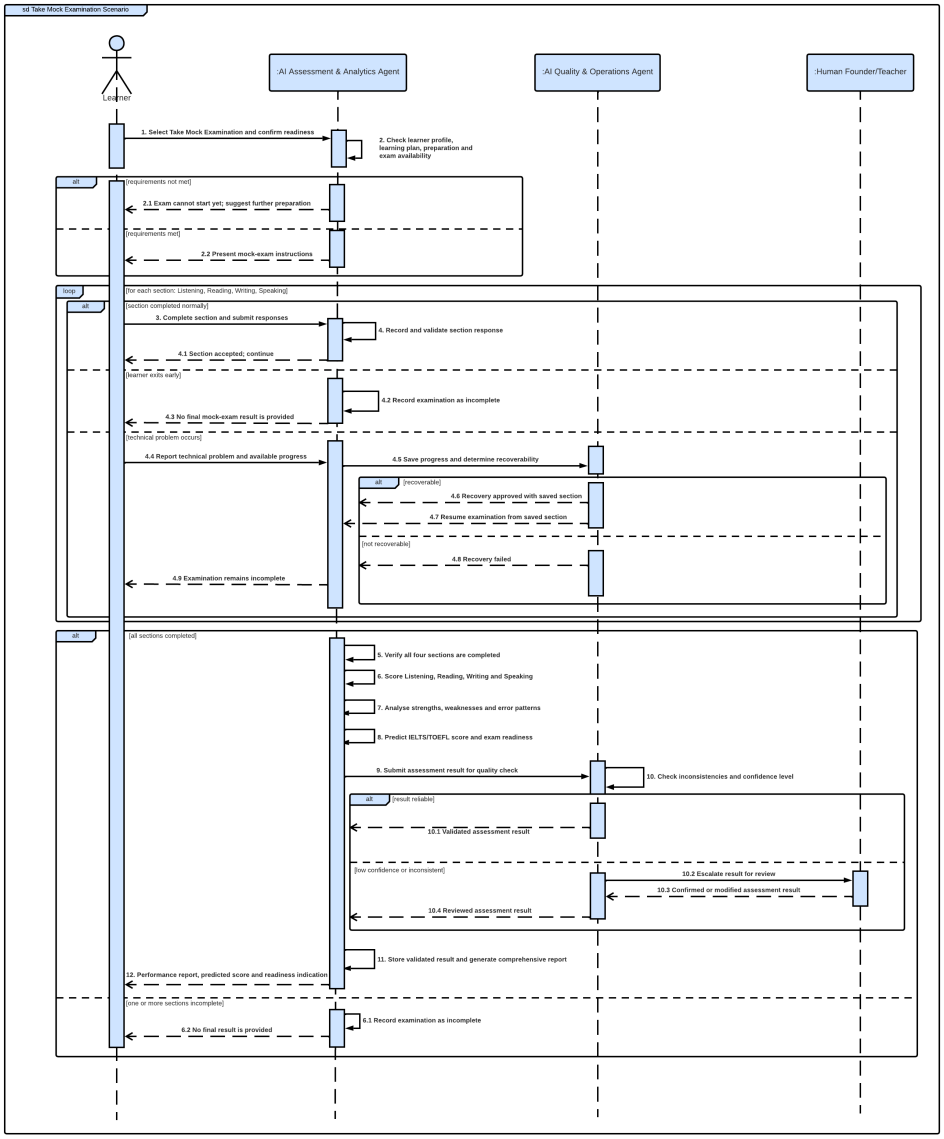


Figure 19: UC5 sequence diagram.

6.3.3 UC5 State Machine Diagram

Figure 20 shows the lifecycle of a mock examination session. The session moves from *Ready to Start* to *Checking Preparation*, where a choice determines whether the examination can proceed or whether further preparation is required. The main examination is represented as the composite state *Exam In Progress*, containing the Listening, Reading, Writing, and Speaking section states. A history pseudostate allows a recoverable interrupted examination to resume from the previously active section. Technical problems may move the session to *Paused / Recovering*, while learner exit or unrecoverable failure leads to *Examination Incomplete*. After successful section completion, the session progresses through *Scoring*, *Performance Analysis*, *Readiness Prediction*, and *Quality Checking*. Reliable results proceed directly to validation, while low-confidence or inconsistent results enter *Human Review* before the report is generated and made available to the learner.

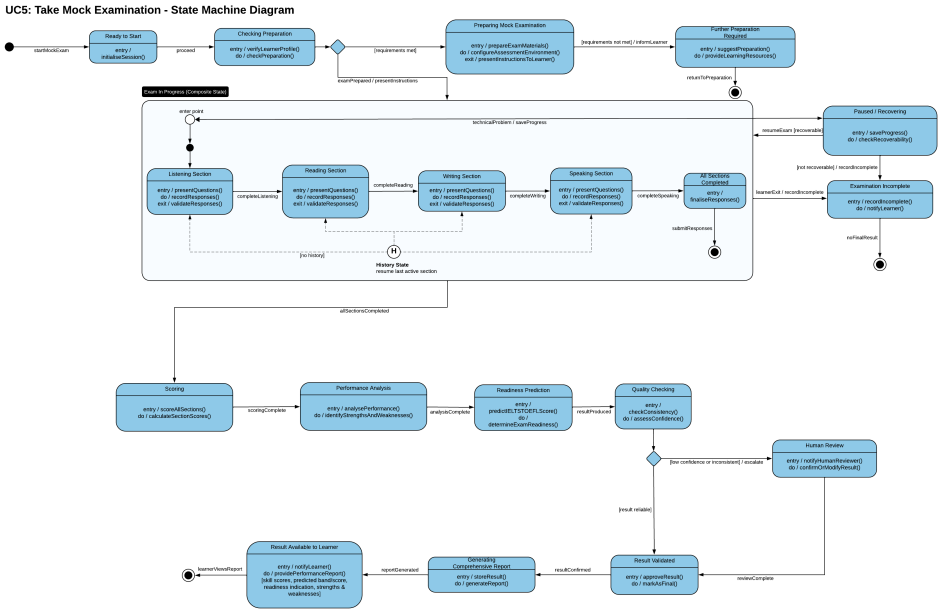


Figure 20: UC5 state machine diagram.

6.3.4 UC6 Activity Diagram

Figure 21 shows the activity flow for viewing a performance report and identifying weaknesses. The learner opens the performance report or progress dashboard, and the assessment agent retrieves available practice and assessment results. A decision separates sufficient data from insufficient data and data-retrieval failure. When sufficient data is available, the assessment agent analyses performance across the four skills, identifies strengths, weaknesses, and repeated problem areas, reviews progress, and generates recommendations. The quality agent then checks the report for inconsistent or low-confidence information. Reliable information proceeds directly to report generation, while flagged information may be resolved by quality review or escalated to the *Human*

Founder/Teacher. The completed report is displayed to the learner, while a display failure produces an unavailable-report path and retains the existing assessment data.

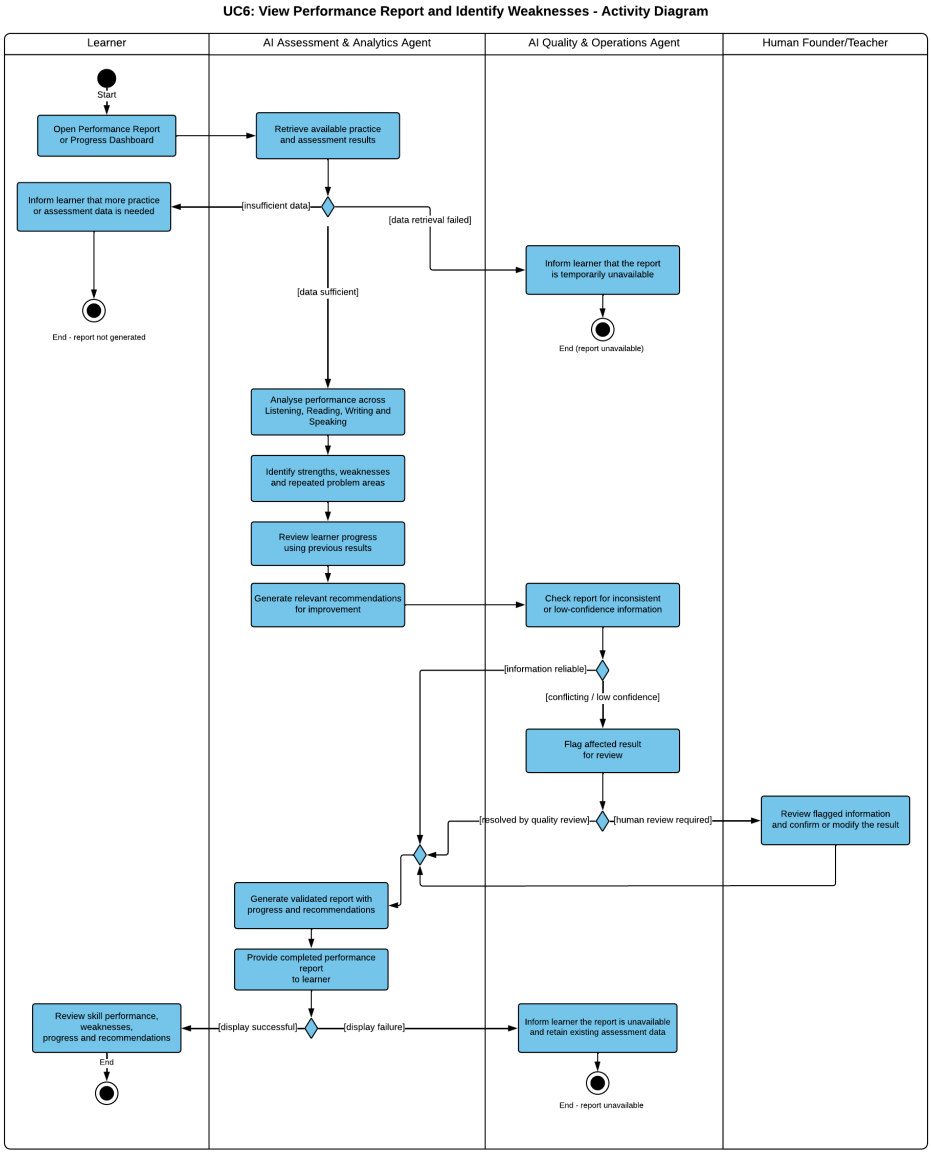


Figure 21: UC6 activity diagram.

6.3.5 UC6 Sequence Diagram

Figure 22 shows the messages between the *Learner*, *AI Assessment & Analytics Agent*, *AI Quality & Operations Agent*, and *Human Founder/Teacher*. The learner first requests the performance report, after which the assessment agent retrieves the available practice, assessment, and mock-examination results. The first **alt** fragment covers insufficient assessment data, retrieval failure, and the normal data-available path. The assessment agent then analyses skill performance, identifies strengths and weaknesses, reviews progress, and generates improvement recommendations. The report is submitted

to the quality agent, where another **alt** distinguishes reliable information from conflicting or low-confidence information. A nested **alt** further separates issues resolved by the quality agent from cases requiring human review. After validation, the assessment agent generates and stores the performance report. The final **alt** represents successful display to the learner or display failure, where the existing assessment data is retained for a later attempt.

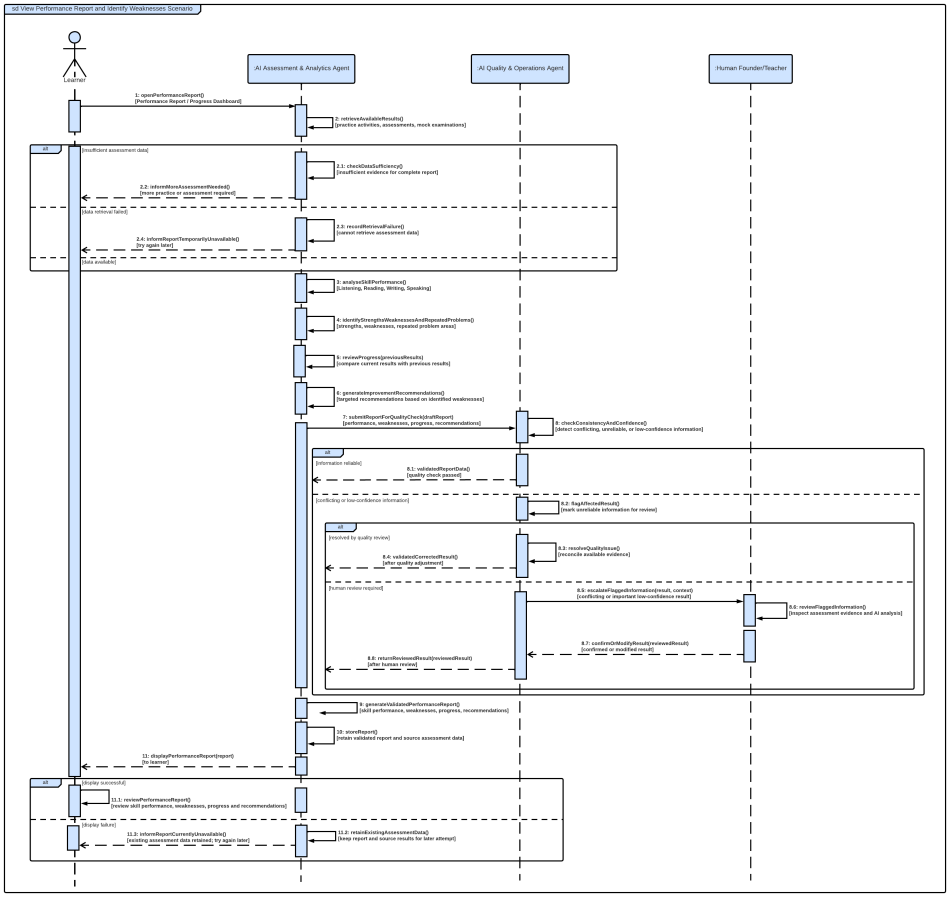


Figure 22: UC6 sequence diagram.

6.3.6 UC6 State Machine Diagram

Figure 23 shows the state transitions involved in generating and displaying the learner's performance report. The process starts in *Ready to View Report* and moves to *Retrieving Performance Data*. Data-availability outcomes include insufficient data and report-unavailable conditions, while the normal path enters the composite state *Performance Report Generation*. Within this composite state, the system moves through *Analyse Four Skills*, *Identify Strengths & Weaknesses*, *Review Progress*, and *Generate Recommendations*. The generated report then enters *Quality Checking*. Reliable reports move to *Report Validated*, while low-confidence or conflicting results enter *Human Review* before validation. The report is then displayed to the learner. A successful display reaches

Report Available to Learner, whereas display failure enters *Display Failure*, where retry is represented by a self-transition and an unrecoverable failure ends with the report unavailable.

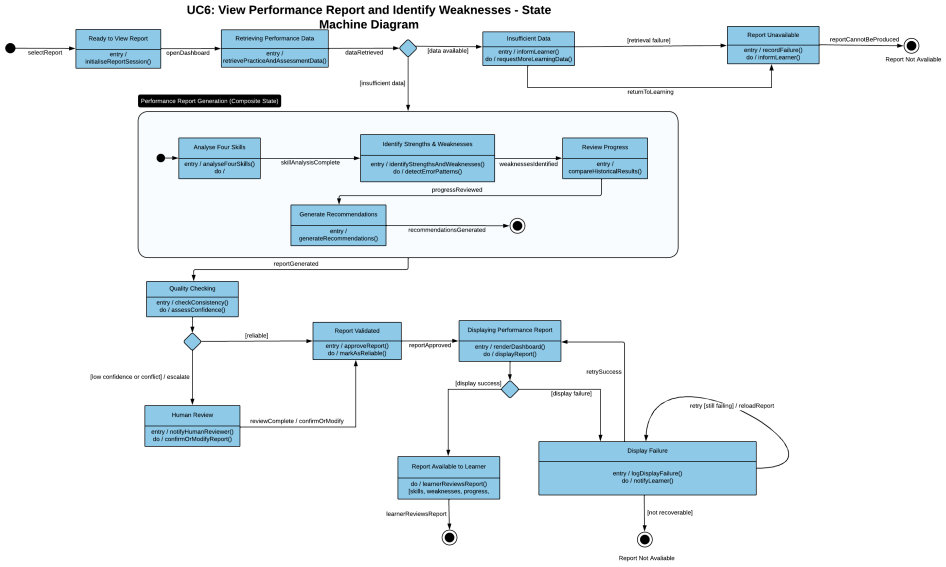


Figure 23: UC6 state machine diagram.

6.4 UC7 and UC8 (Mohamed Azim)

6.4.1 UC7 Activity Diagram

Figure 24 shows the UC7 workflow across four swimlanes: *Learner*, *System*, *Learning Coordinator* and *LLM Service*. After a trigger, the coordinator compares recent progress with the study priorities. If there is not enough evidence, the existing plan is kept and the flow ends. Otherwise the LLM proposes an adapted plan (with one retry), resources are selected, and the plan is presented to the learner. If the learner rejects it, the rejection is recorded and the existing plan stays active. If the learner confirms, the plan is activated and the other agents are notified, with retries on failure. Each way the use case can end has its own final node.

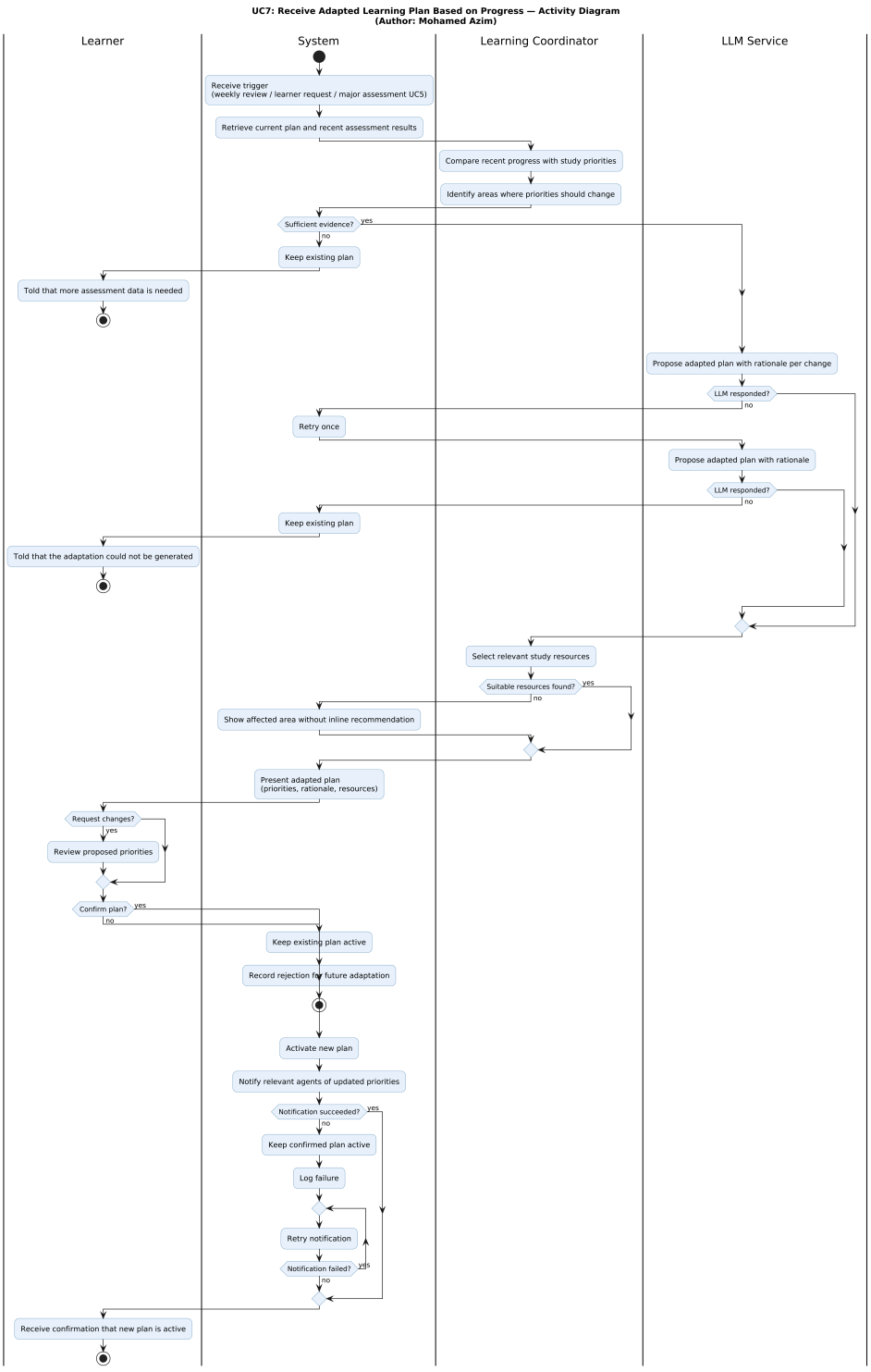


Figure 24: UC7 activity diagram.

6.4.2 UC7 Sequence Diagram

Figure 25 shows the messages between the *Learner*, *System*, *Learning Coordinator*, *LLM Service* and *Other Agents*. The system compares the current plan with recent assessments through the coordinator. The outer **alt** fragment handles insufficient evidence (3a). Otherwise the LLM proposes an adapted plan, with an **opt** retry if it does not respond (4a).

The coordinator then selects resources, and the plan is shown to the learner, who may request changes (6a). A further **alt** covers confirmation and rejection (7a). On confirmation the plan is activated and the other agents are notified, with a **loop** that retries a failed notification (8a).

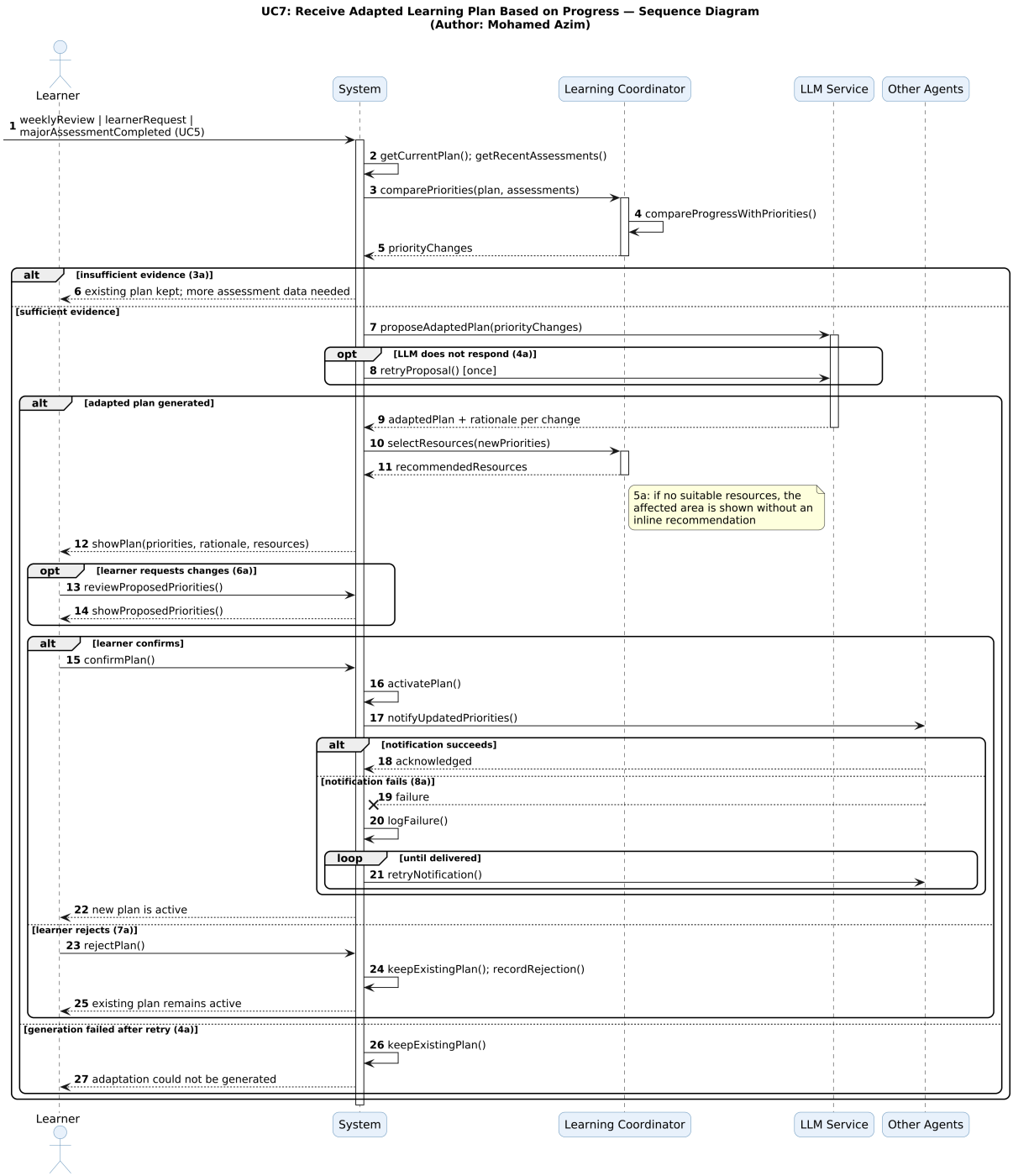


Figure 25: UC7 sequence diagram.

6.4.3 UC7 State Machine Diagram

Figure 26 follows an *AdaptedPlan* through its lifecycle, from *Triggered* and *Comparing* to *Proposing*, *SelectingResources* and *Presented*. From *Presented*, the learner can confirm the plan (*Activating*), request changes (*ReviewingChanges*) or reject it (*Rejected*). Confirmation leads through *Notifying*, with a *NotificationRetry* loop on failure, to *Active*. The failure states *InsufficientData*, *GenerationFailed* and *Rejected* sit inside the compound state *KeptExisting*, which keeps the existing plan. Transitions use the *trigger [guard] / effect* notation, with time events and entry actions shown inside the states.

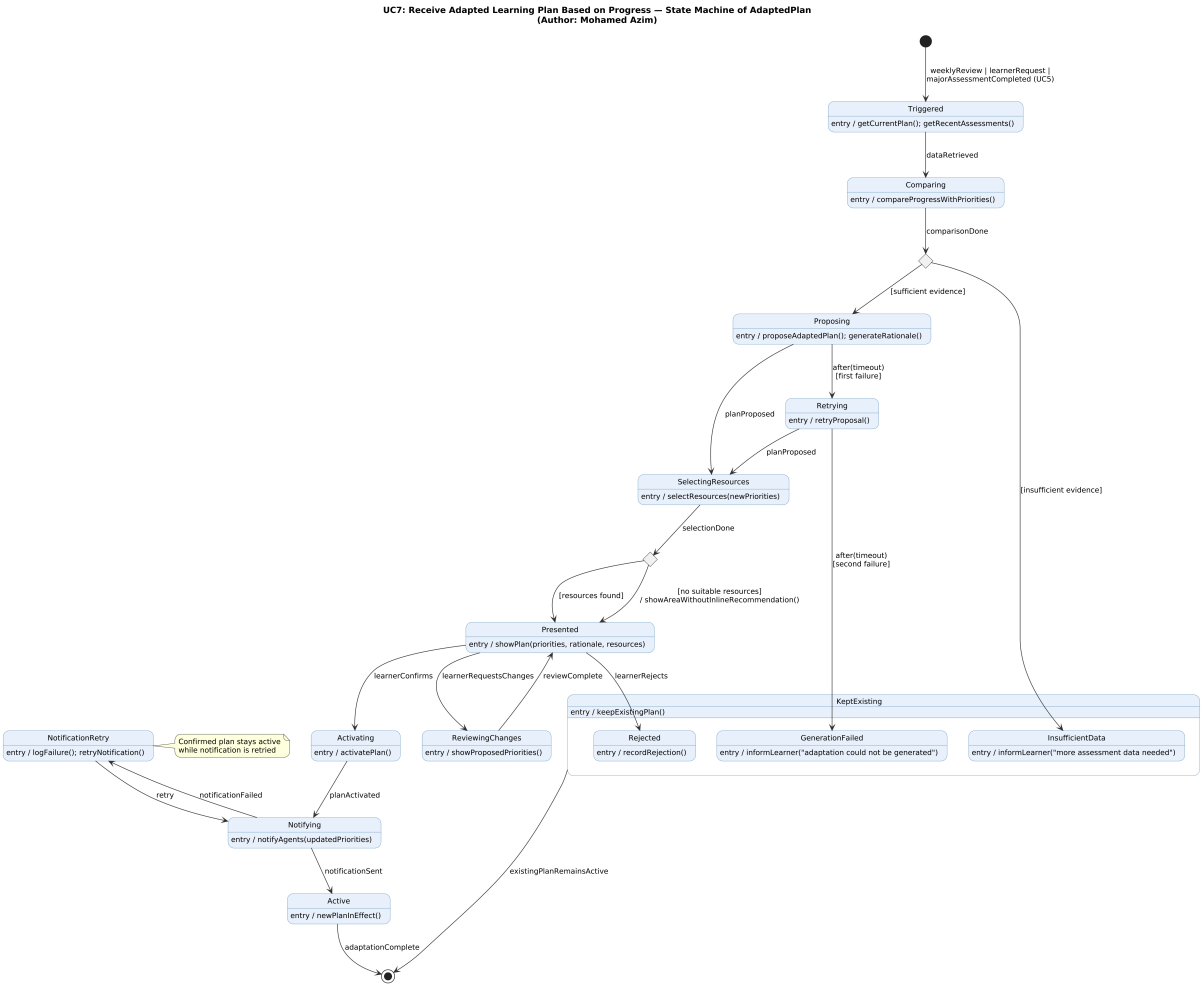


Figure 26: UC7 state machine diagram (AdaptedPlan).

6.4.4 UC8 Activity Diagram

Figure 27 shows the UC8 workflow across three swimlanes: *Quality & Operations Agent*, *System* and *Human Instructor*. The agent creates an escalation, and the system notifies the instructor, queuing the case if no instructor is available. The instructor reviews the case evidence, with conflicts and low-confidence results highlighted by the system. If there is not enough context, the case stays open and more information is requested.

Otherwise the instructor approves the AI result, overrides it with a new decision, or requests coaching or more information. Once a decision is made, the system records it with the reasoning and notifies the relevant agents, retrying on failure. The case is then marked resolved and made available for future quality checks.

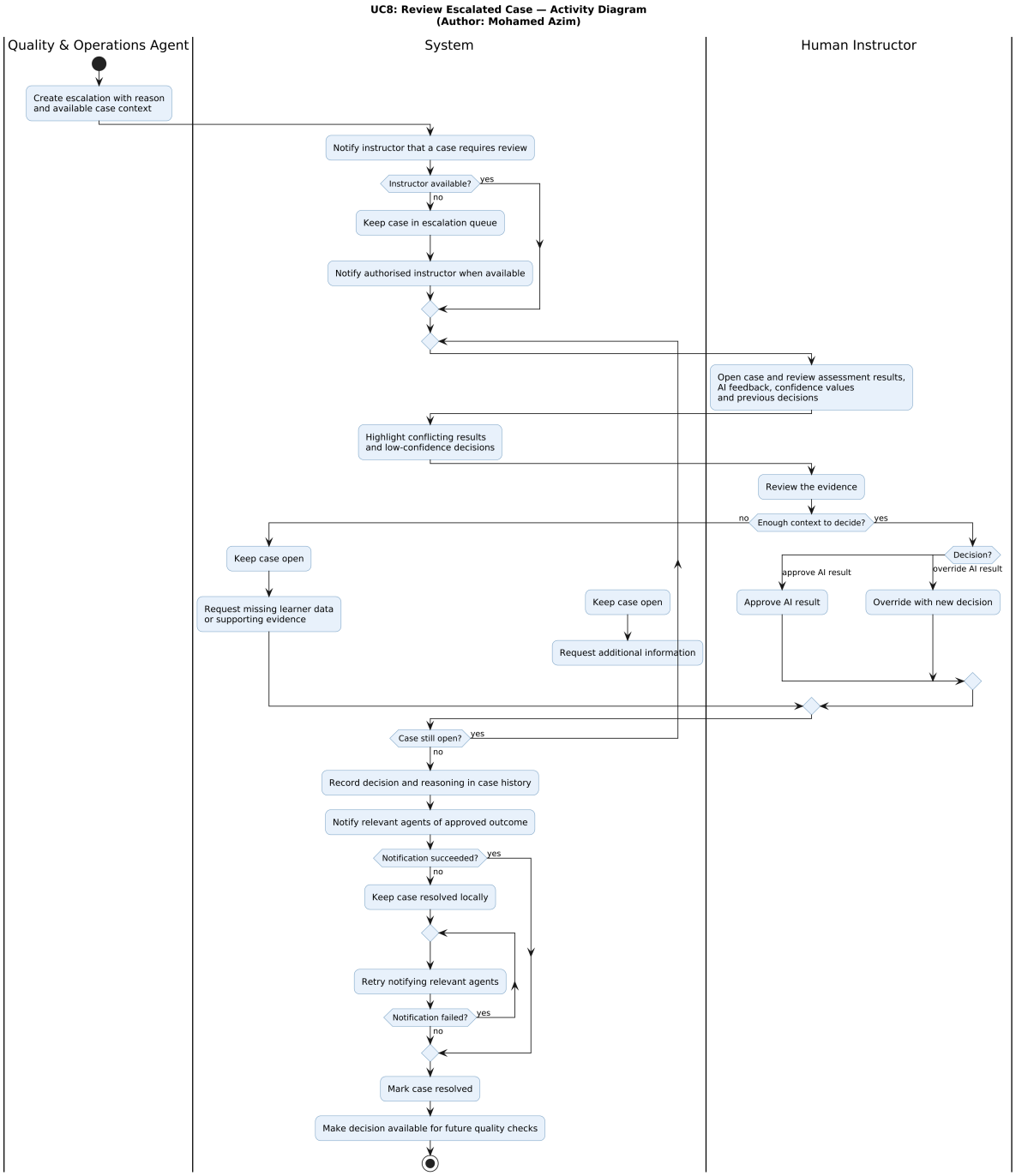


Figure 27: UC8 activity diagram.

6.4.5 UC8 Sequence Diagram

Figure 28 shows the messages between the *Quality & Operations Agent*, *System*, *Human Instructor* and *Relevant Agents*. The agent creates the escalation, the system notifies the instructor, and an **opt** fragment keeps the case in a queue if the instructor is unavailable (2a). The instructor opens the case and receives the case details with the conflicts highlighted. A second **opt** requests missing evidence from the agent if the context is insufficient (3a). An **alt** fragment then separates requesting coaching or more information (5c) from making a decision, which is either approving (5a) or overriding (5b) the AI result. After the decision is recorded, the relevant agents are notified, and a **loop** retries a failed notification until it is delivered (7a). The case is then marked resolved.

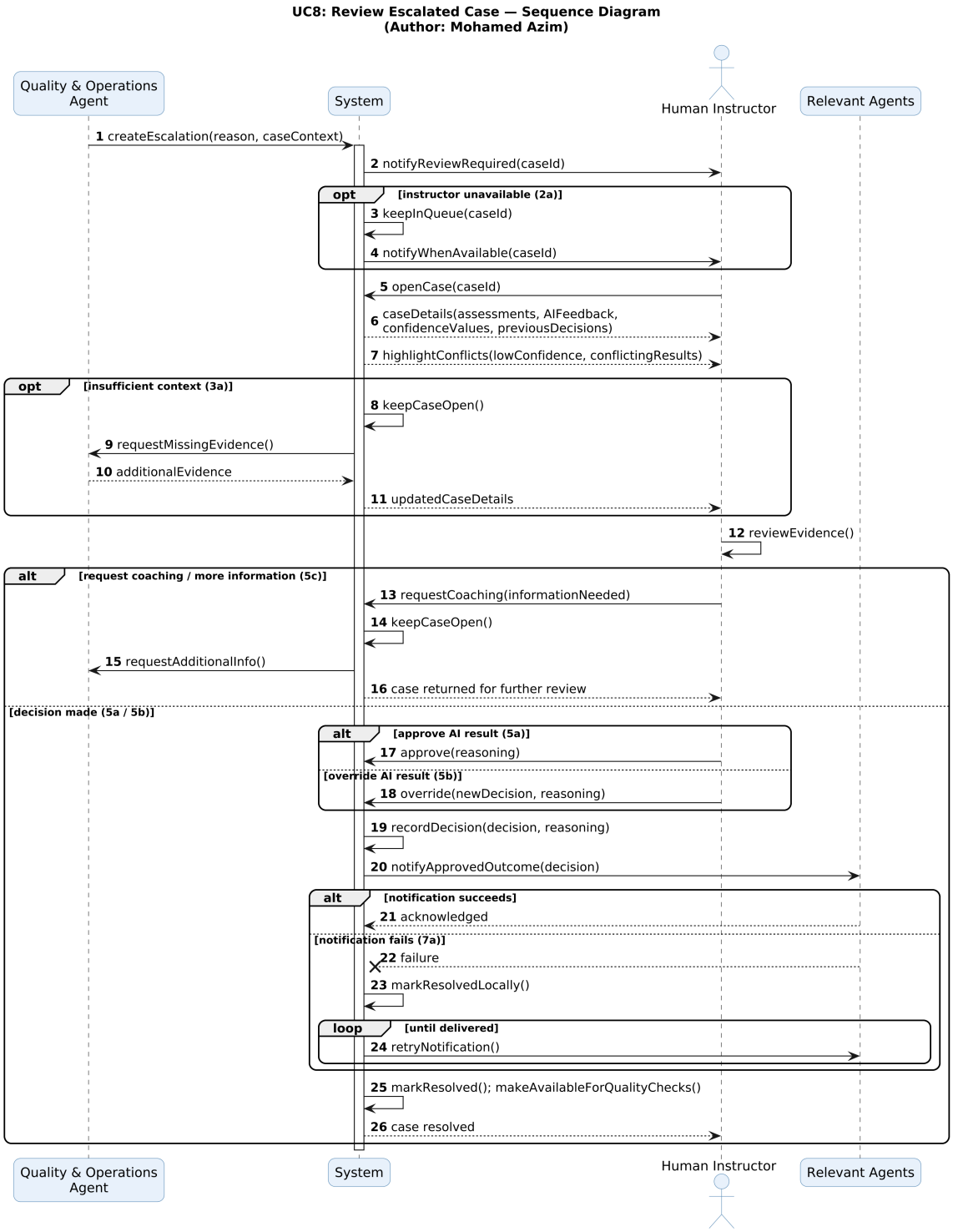


Figure 28: UC8 sequence diagram.

6.4.6 UC8 State Machine Diagram

Figure 29 follows an *EscalatedCase* through its lifecycle. The case moves from *Escalated* to *NotifyingInstructor*, then through a choice pseudostate: *UnderReview* if an instructor is available, or *Queued* if not. *Queued*, *UnderReview* and *AwaitingInformation* sit inside the compound state *Open*, so the case stays open until a decision is made. After a

decision (approve or override), the case moves to *Decided* and then *NotifyingAgents*, with a *NotificationRetry* loop on failure, before reaching *Resolved* and the final state. Transitions use the *trigger [guard] / effect* notation, with entry actions shown inside the states.

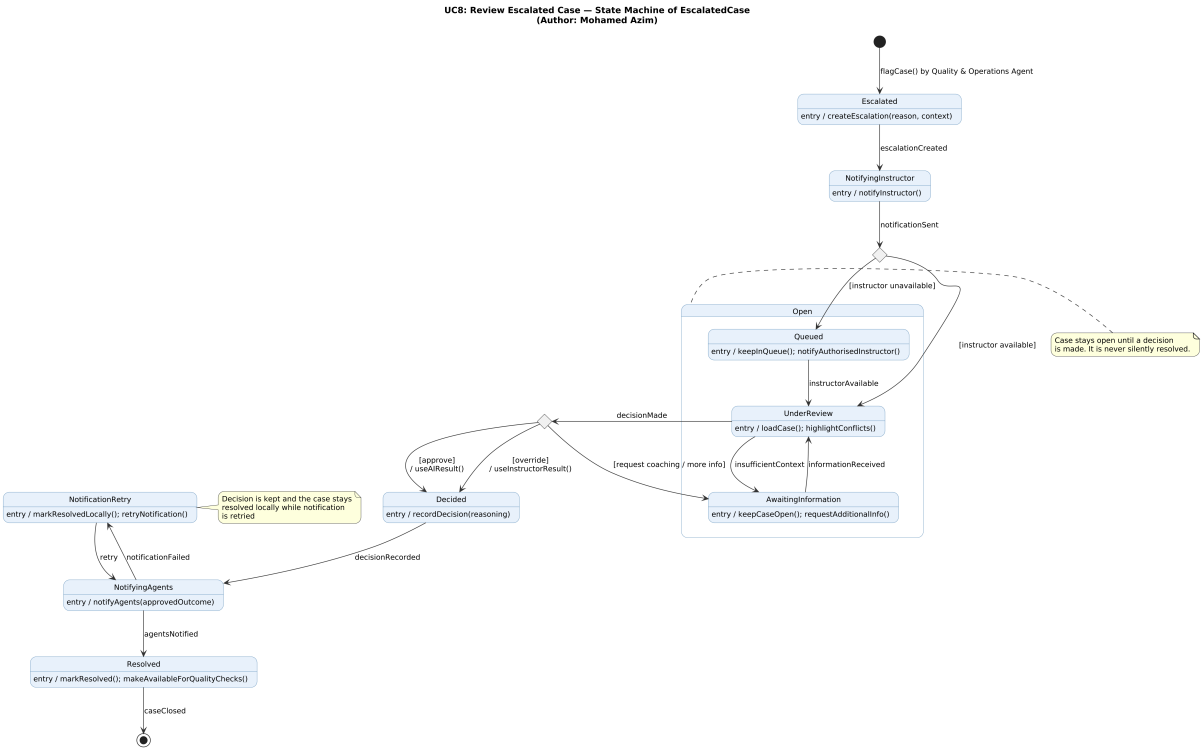


Figure 29: UC8 state machine diagram (EscalatedCase).

6.5 UC3 and UC4

6.5.1 UC3: Complete Speaking Practice Session

6.5.2 Activity Diagram

Figure 30 shows the activity flow for UC3, Complete Speaking Practice Session. The learner selects speaking practice, receives a speaking task, prepares and records a response, and submits the recording. The system analyses the speaking performance and generates feedback and an assessment result. The result confidence and consistency are then checked before the feedback is released. Low-confidence or inconsistent results are held and escalated to the human instructor for review.

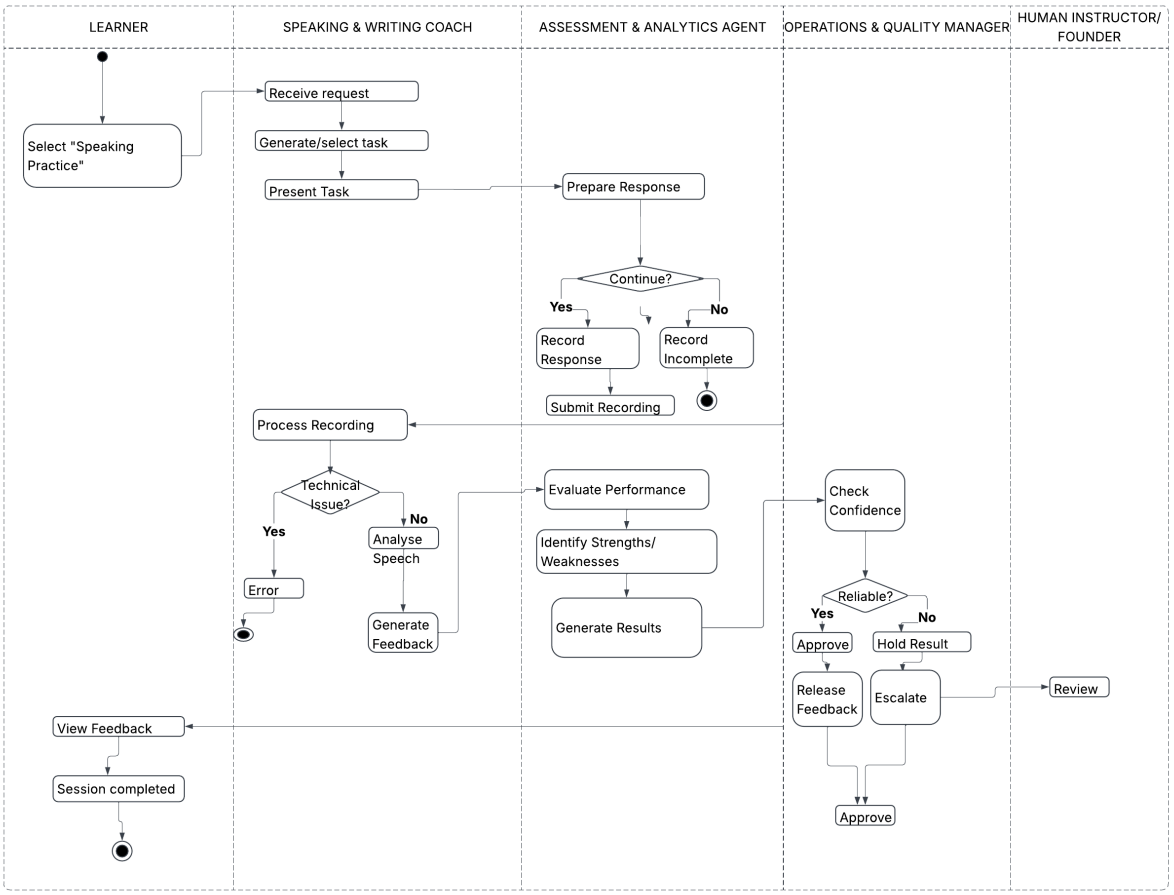


Figure 30: UC3 activity diagram.

6.5.3 Sequence Diagram

Figure 31 shows the interaction flow for UC3 between the learner, Speaking & Writing Coach, Assessment & Analytics Agent, Operations & Quality Manager, and Human Instructor. The sequence covers task generation, speaking submission, performance assessment, confidence checking, and the escalation of unreliable results.

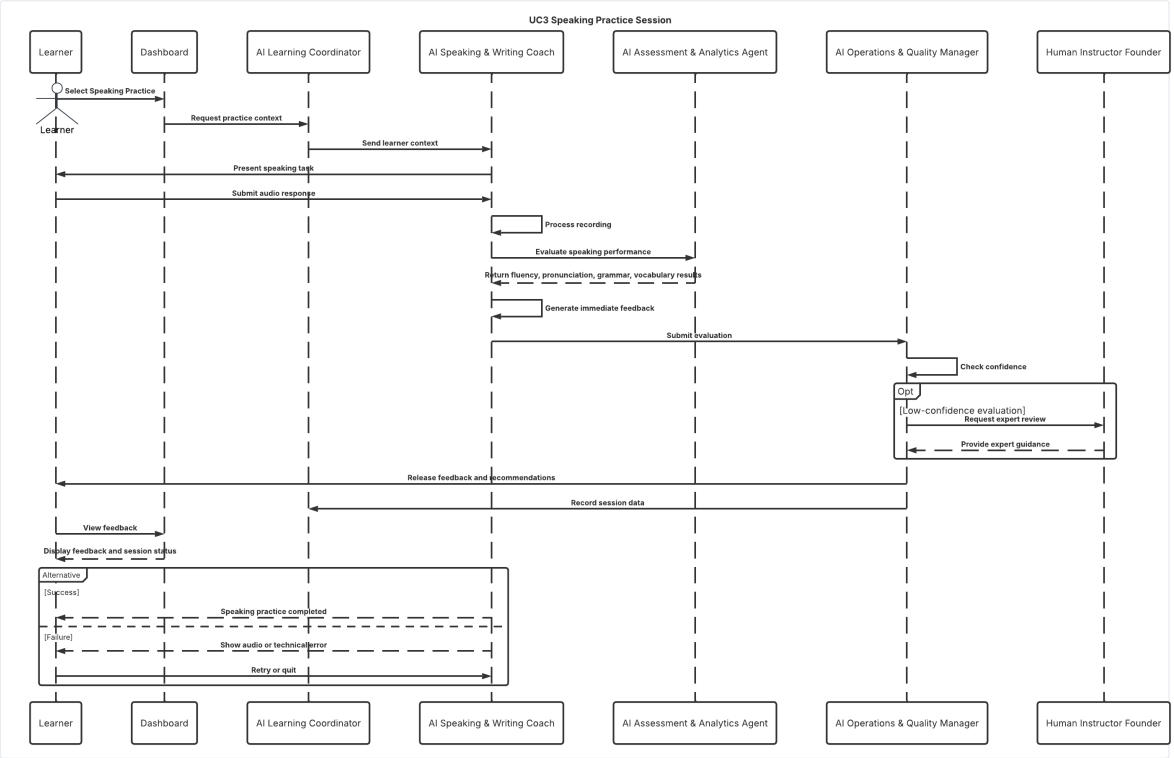


Figure 31: UC3 sequence diagram.

6.5.4 State Machine Diagram

Figure 32 shows the state transitions for the speaking practice session. The session progresses from task selection and response recording through assessment and feedback generation. A reliable result can be released to the learner, while a low-confidence or inconsistent result is held for human review before release.

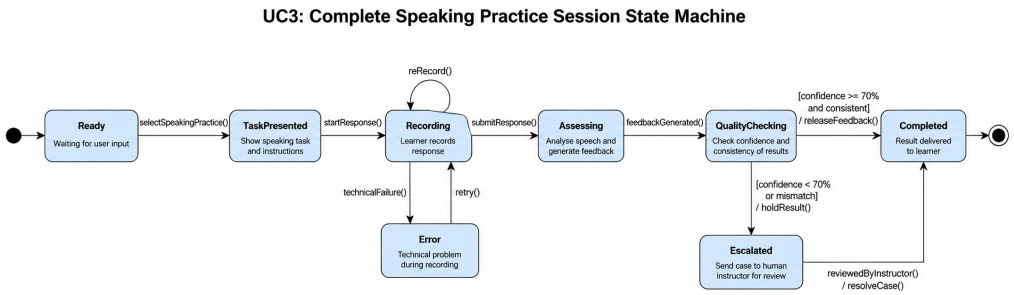


Figure 32: UC3 state machine diagram.

6.5.5 UC4: Complete Writing Practice with Feedback

6.5.6 Activity Diagram

Figure 33 shows the activity flow for UC4, Complete Writing Practice with Feedback. The learner selects writing practice, receives a writing task, submits a response, and the system assesses the writing performance and generates feedback. The result is checked before being released to the learner, with problematic results escalated for human review when required.

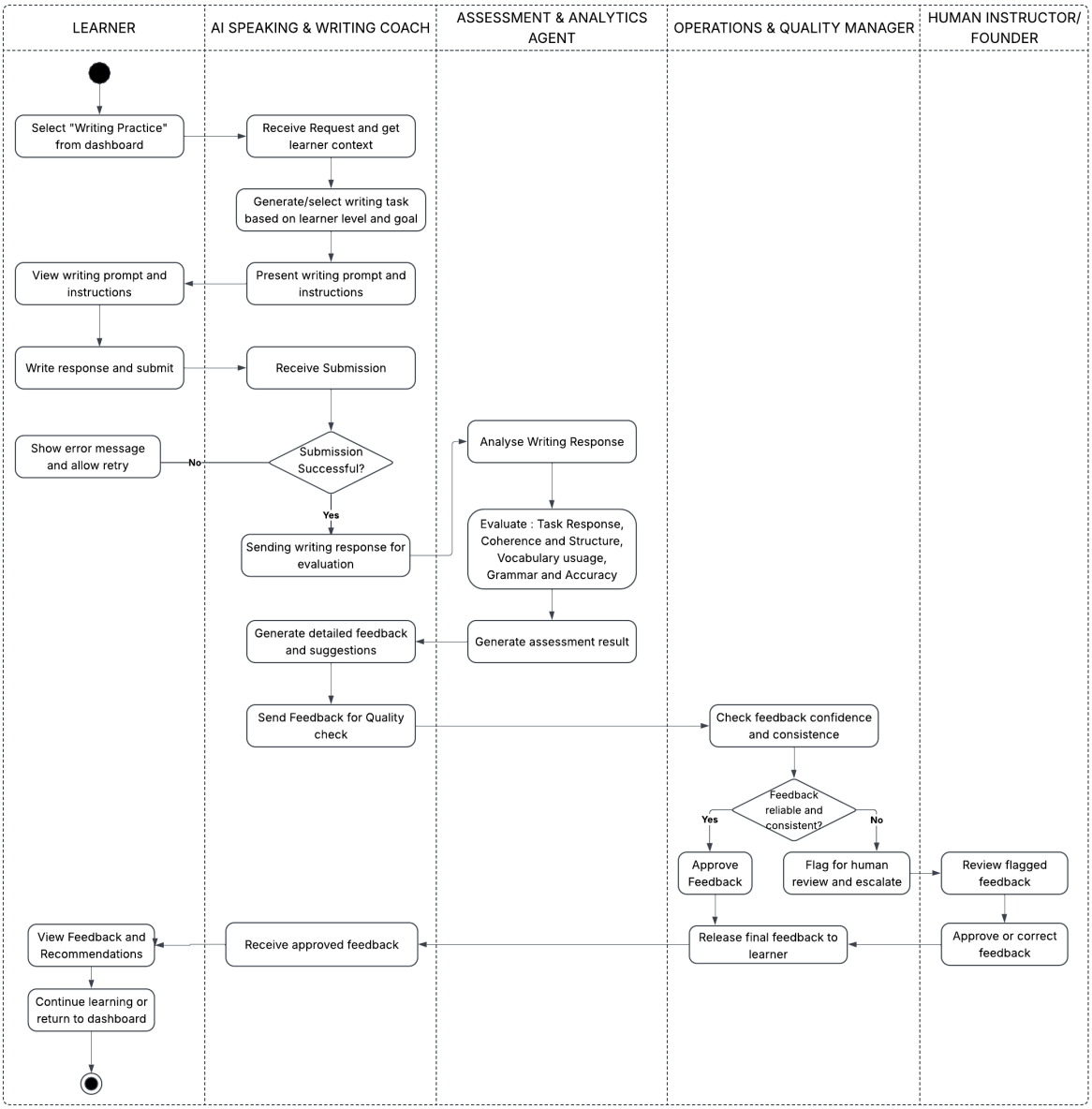


Figure 33: UC4 activity diagram.

6.5.7 Sequence Diagram

Figure 34 shows the interaction flow for UC4 between the learner, Speaking & Writing Coach, Assessment & Analytics Agent, Operations & Quality Manager, and Human Instructor. The sequence covers writing task generation, response submission, assessment, feedback generation, quality checking, and result delivery.

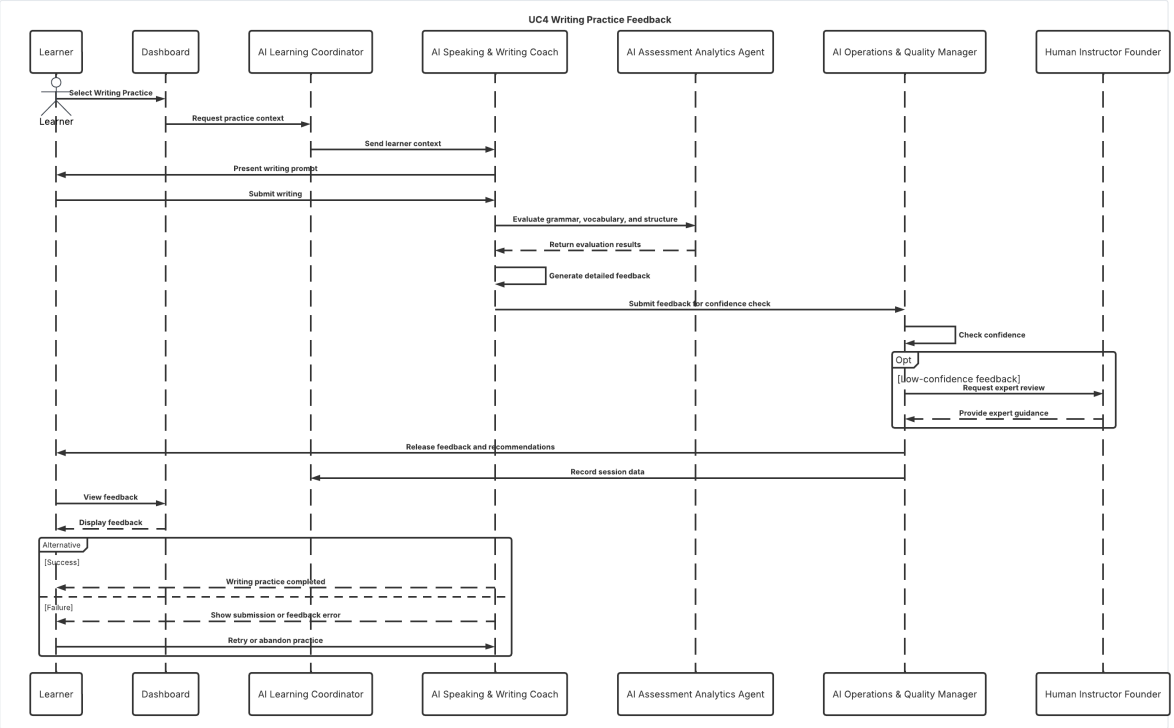


Figure 34: UC4 sequence diagram.

6.5.8 State Machine Diagram

Figure 35 shows the state transitions for the writing practice session from task selection and response submission through assessment and feedback generation. The result is released when the quality checks are satisfied; otherwise, the result is held for further review.

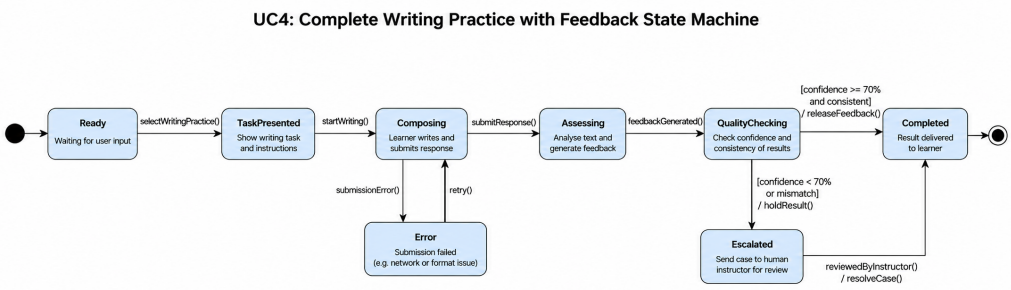


Figure 35: UC4 state machine diagram.

7 Role of the LLM in the Agent System

An LLM on its own only answers a prompt; it becomes part of an agent when it works inside a goal-directed loop of perceiving, deciding, acting and evaluating results. Table 21 shows where the LLM contributes to each agent.

Agent	**Perception**	**Decision-making**	**Action**
Learning Coordinator	Interprets the learner's goals and profile	Prioritises skills when creating or adapting plans; ranks materials	Explains the plan; writes reasons for recommendations; generates targeted exercises
Tutor	Reads essays and transcribed speech	Identifies errors and their severity	Writes prompts, feedback and corrected examples
Assessment & Analytics	Reads responses against the band descriptors	Assigns bands with confidence values; detects error patterns	Produces reports and readiness predictions
Operations & Quality Manager	Reads the outputs of the other agents	Judges whether feedback and scores contradict each other	Releases the result or holds and escalates it

Table 21: Contribution of the LLM to each agent.

Because LLMs can be confidently wrong, the design includes these safeguards:

- structured output with confidence values (*LLMResponse*, *Verdict*);
- cross-agent consistency checking and a confidence threshold (UC9);
- human review of escalated cases (UC8);
- retry and rule-based fallback when the LLM is unavailable;
- removal of personal details before requests are sent;
- audit logging of every check and decision.

8 Assumptions

- The LLM can return structured output (JSON) containing scores, errors and confidence values.
- The confidence values reported by the LLM are reasonably calibrated; the threshold (default 0.7) is set and adjusted by the instructor.

- The instructor reviews escalated cases within 24 hours.
- Learners have a stable internet connection and a microphone for speaking practice.
- Official IELTS/TOEFL band descriptors can be used as the scoring rubric.
- Each learner prepares for one target exam at a time.
- The system is a preparation aid; it does not award official scores.

9 Change Assessment and Acceptance Determination

9.1 Acceptance Criteria

The system is accepted when the measurable targets in Table 22 are met in testing.

Criterion	**Target**	**How it is measured**
Feedback speed	First feedback within 3 s	Timed test submissions
Scoring accuracy	AI band within ±0.5 of an examiner for at least 85% of essays	Sample of essays marked by the instructor
Scoring consistency	Inter-rater reliability above 0.85	Re-score the same responses
Quality check	False positives below 2%; false negatives below 1%	Instructor-labelled sample of results
Availability	99.9% uptime	Monitoring over a trial period
Usefulness	More than 85% of learners rate feedback helpful	Learner survey

Table 22: Acceptance criteria.

9.2 Change Assessment

Proposed changes (e.g. a new prompt, a new LLM model or a new threshold) are assessed by:

1. identifying the affected features in the feature diagram, and following cross-tree constraints to find dependent features;
2. identifying the affected use cases and classes through the include/extend relationships and the class associations;
3. re-running the acceptance tests in Table 22 on the same labelled sample;
4. comparing the escalation and contradiction rates recorded in the *AuditLog* before and after the change, and rolling back if they increase.

10 Individual Contributions

Member	Contribution	Share
Naratorn Pisedtasalasai	Ad hoc requirement (essay feedback); use case specifications UC9 and UC10; activity, sequence and state machine diagrams for UC9 and UC10; revised overall use case diagram; revised feature diagram; class diagram; object and collaboration diagrams.	22%
Menglu Lin	Ad hoc requirement, use case specifications UC1 and UC2; activity, sequence and state machine diagrams for UC1 and UC2; checked the overall use case diagram; refinement and consistency checking of the learner profile and personalised learning plan models.	20%
Yuxuan Tang	Ad hoc requirement, use case specifications UC5 and UC6; activity, sequence and state machine diagrams for UC5 and UC6; Ad hoc diagram; Object diagram draft.	18%
Mohamed Azim	Ad hoc requirement, use case specifications UC7 and UC8; activity, sequence and state machine diagrams for UC7 and UC8; feature diagram; revised and checked class diagram; revised and checked object and collaboration diagrams along with Naratorn.	20%
Suhani Saha	Ad hoc requirement, use case specifications UC3 and UC4; activity, sequence and state machine diagrams for UC3 and UC4; Combine the group videos.	20%

Table 23: Individual contributions.

11 YouTube Video Link

https://youtu.be/YBu4lVqm0vs

References

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